



Quantitative methods

—

Advanced topics and a primer to SPSS

Welcome!



Aim of the session – prepare you for assignment 3

Prep. Decide on a main objective/RQ, get your data into SPSS and familiarize yourself with the survey and codebook

Part A

Exploratory analysis of the data (e.g. frequencies and tables)

Reliability test on a construct (perceived stress)

Part B

Correlations

Part C

Regression

**Remember to interpret your results and
reference methods literature!**

**Seminar: 12 October
Submission: 16 October**



Agenda

Part 1

Advanced topics

Part 2

A primer to SPSS



Remember?

Question: How would you measure your intelligence?

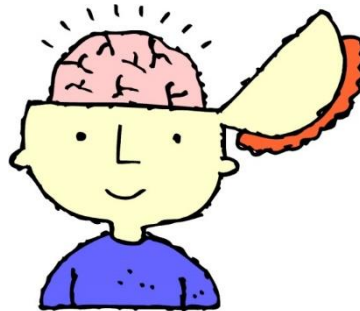
Remember...?



Creation of constructs

As a general rule: Multiple ways of measuring something is preferable since it offers a better/multifaceted representation of the construct.

Question: Would you feel comfortable with someone measuring your intelligence with only one question?



Construct creation

Many ways of creating a construct – average, median, sum of a set of variables. But there are more sophisticated ways:

Exploratory factor analysis is often used interchangeably with **Principal component analysis**.

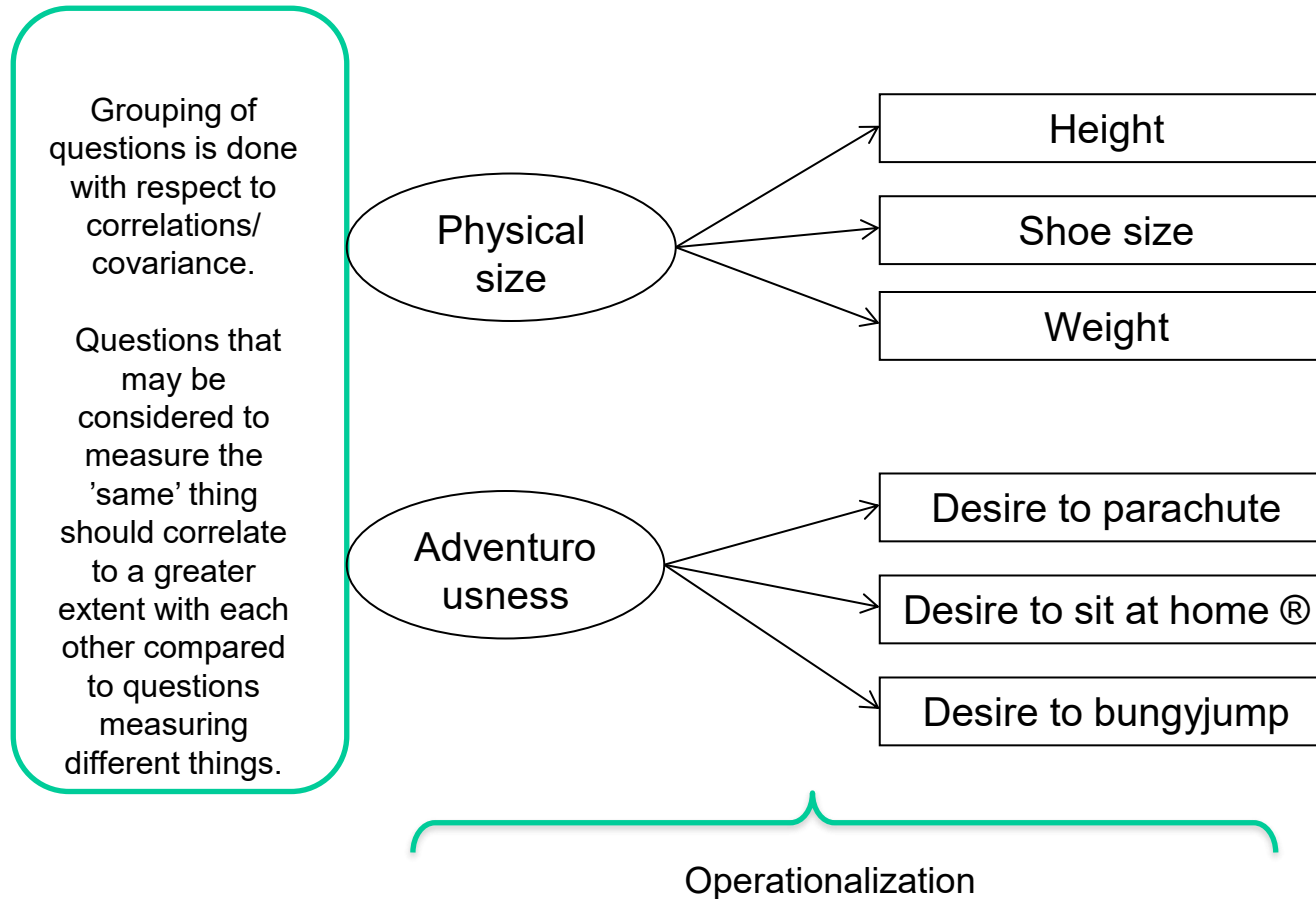
- Considers inter-relationships between variables.

- Input of many variables will be factored (divided) into groups based on common underlying dimensions (existing within the data).

- Allows you to explore dimensions of your data, or confirm a structure in the data.

Confirmatory factor analysis – theory testing.

How do we know which questions reflect a construct?



Independence of constructs

Consider the example of having two constructs you are using as independent variables in your study, **Contributory role** and **Specialized resources**.

Contributory Role. Please estimate the approximate percentage of your subsidiary company's revenues that are gained as a result of your international responsibilities.

1-7 scale

Specialized resources. Indicate your capability or distinctive expertise in the following areas relative to other subsidiaries in the corporation:

- (1) product or process R&D;
- (2) manufacturing capability;
- (3) marketing capability;
- (4) managing international activities;
- (5) innovation and entrepreneurship.

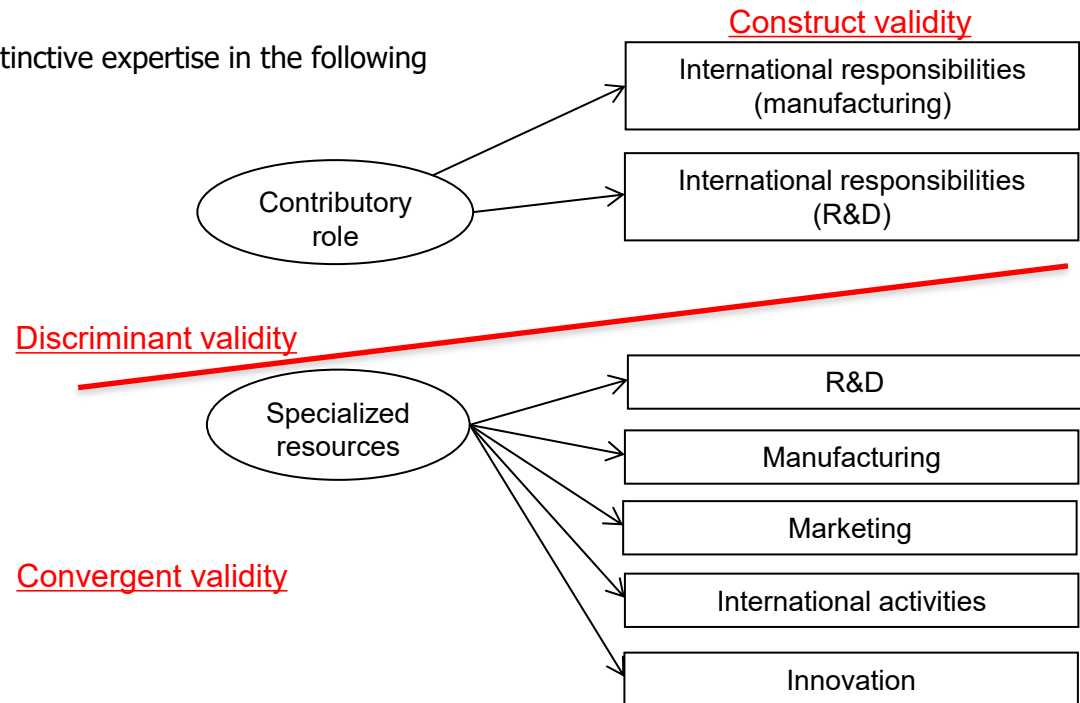
1 = far below average, 7 = far above average.

Birkinshaw, Hood & Jonsson (1998)

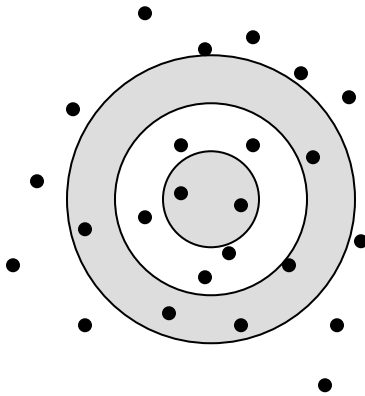
Rotated Component Matrix^a

	Component		
	1	2	3
RESEARCH	.875	.436	-.209
FIELD	.780	.256	.572
WITHIN	.568	-.674	-.472
WELL	.568	-.674	-.472
LEVEL	.332		-.940
COMMUNICATION	-.147	.968	.202
APPLIED	.533	.843	
AREA	.483	.841	.243
PUBLIC	.345	.766	.542
THEORETIC	.345	.766	.542
RELATION	.345	.766	.542
TOOLS	.345	.766	.542
CONCEPTUAL	.345	.766	.542
DEVELOPED	.585	.734	-.344
CONCLUDES	-.332		.940
YEARS	.579		.811
ACROSS	.579		.811
APPLY	.579		.811
BASED	.579		.811
TRENDS	.579		.811
VISUAL	.324	-.860	.395
STUDIES	-.343	-.852	.395

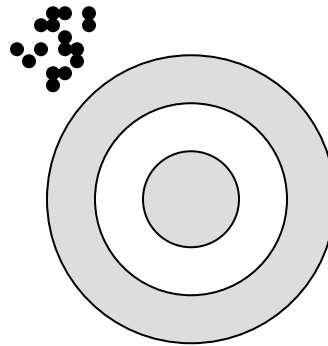
Extraction Method: Principal Component Analysis.
Rotation Method: Varimax with Kaiser Normalization.
a. Rotation converged in 6 iterations.



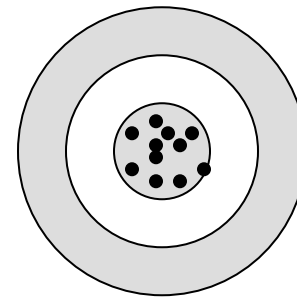
Construct reliability and validity



Nether reliable nor valid



High reliability but no validity



High reliability and validity

- **Reliability:** (**Stability**) Repeated measures of the same thing provides consistent results (think Cronbach Alpha (>0.7 as rule of thumb)).
- **Validity:** You measure what you intend to measure. In factor analysis we look at **convergent** and **discriminant validity**.

Factor analysis

- Thompson, B. (2004). *Exploratory and confirmatory factor analysis: Understanding concepts and applications*. American Psychological Association.
- Comrey, A. L., & Lee, H. B. (2013). *A first course in factor analysis*. Psychology Press.
- Brown, T. A. (2015). *Confirmatory factor analysis for applied research*. Guilford Publications.
- Osborne, J. W., & Costello, A. B. (2009). Best practices in exploratory factor analysis: Four recommendations for getting the most from your analysis. *Pan-Pacific Management Review*, 12(2), 131-146.
- O'Rourke, N., & Hatcher, L. (2013). *A step-by-step approach to using SAS for factor analysis and structural equation modeling*. Sas Institute.
- Jöreskog, K. G., Sörbom, D., & Magidson, J. (1979). Advances in factor analysis and structural equation models.

Questionnaires: Common method bias



Common method bias

Questionnaire example:

Question 1: Do you like your boss?



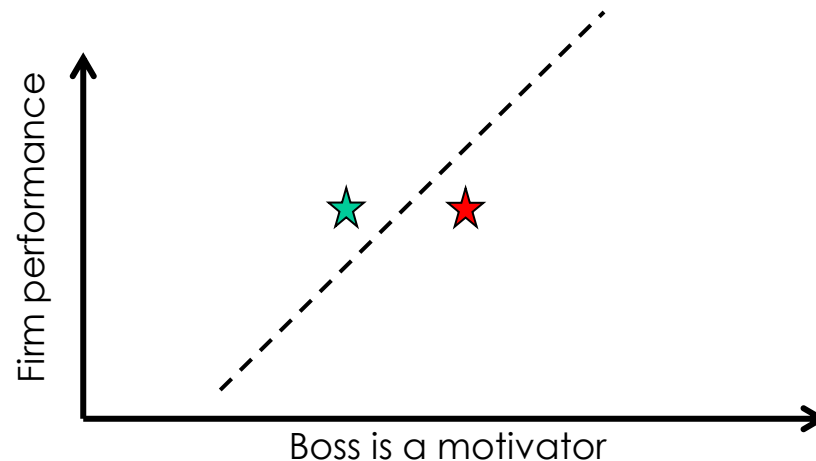
Question 27: Is your boss a good motivator?

Error variance attributed to the measurement method rather than the construct the measures are assumed to represent.

Common method bias

In research: when the **same individual** fills in the values of both independent and dependent variables in a survey (or in an interview setting).

Suppose that some individuals have a "negativity bias", and others a "positivity bias".



Consistency bias may also be at play in this scenario.

Common method bias

So, in sum, common method bias:

Refers to the degree (or magnitude) of bias between the observed and true relationships between constructs that is the result of common method variance.



In quantitative research, this increases the likelihood of finding statistical relationships that are not really there or rejecting relationships that should not have been rejected.

The magnitude of the issue is difficult to judge!



Common method bias

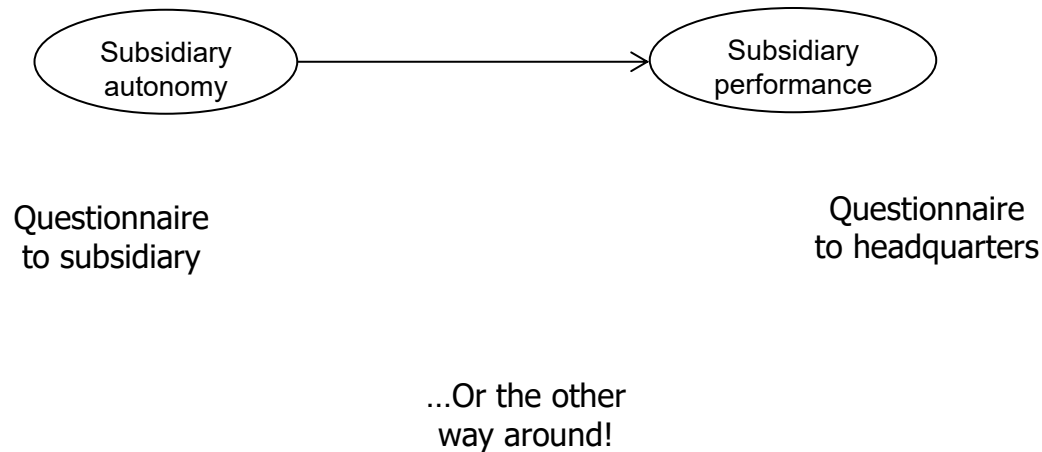
How do we deal with or avoid common method variance / bias?



Common method bias

Remedy 1 example: **Multiple respondents**

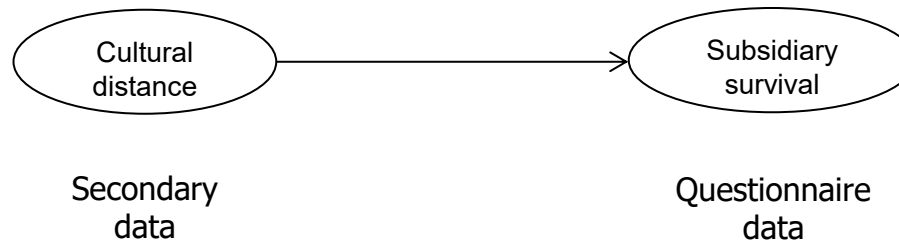
Example: Subsidiary autonomy and its influence on performance.



Common method bias

Remedy 1 example: **Combination of data sources**

Example: Cultural distance between foreign subsidiary and home country of the firm and its influence of performance.



Common method bias

Remedy 2 example: **Mixing scales, question order and sections of the survey**

Contributory Role. Does your subsidiary company have any international responsibilities or world mandates (that is, does it undertake any activity such as manufacturing, R&D or product management on behalf of the corporation as a whole?)

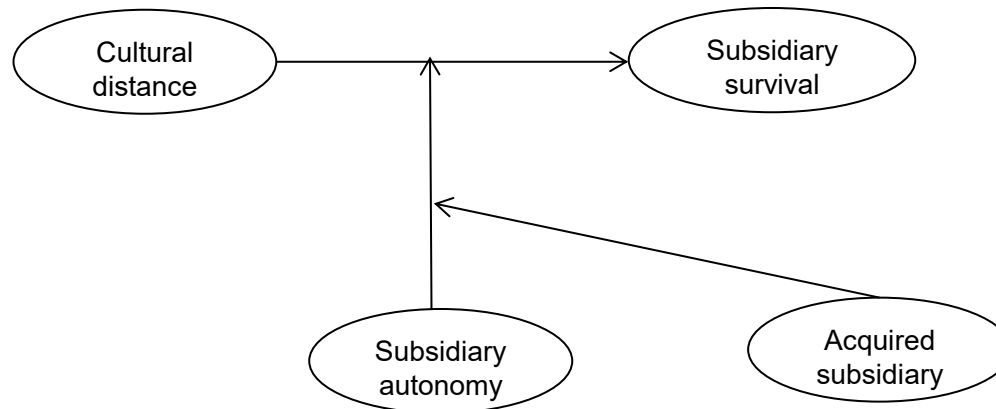
Specialized resources. Indicate your capability or distinctive expertise in the following areas relative to other subsidiaries in the corporation:

- (1) product or process R&D;
 - (2) manufacturing capability;
 - (3) marketing capability;
 - (4) managing international activities;
 - (5) innovation and entrepreneurship.
- 1 = far below average, 7 = far above average.



Common method bias

Remedy 3 example: **Complex model specifications**



Common method bias

Remedy 4 example: **One large Factor analysis**

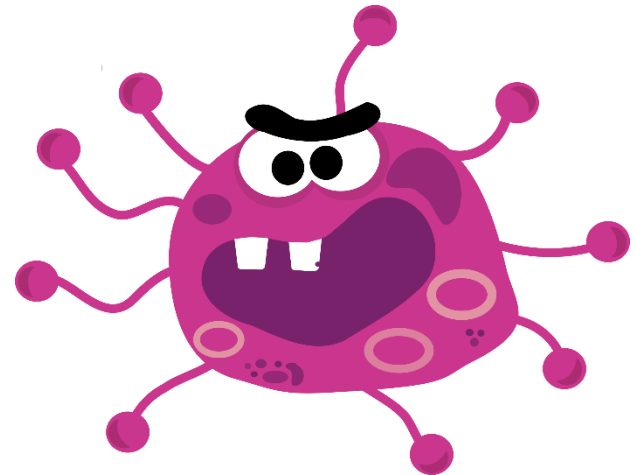
Include all the indicators in a factor analysis and run an exploratory factor analysis (a so called Harman's one factor test) to see if there is a common structure (variance) in your data!

If not – good, but we **cannot rule out common method bias.**

The test has been found to be insensitive, i.e. it is unlikely that you will ever fail a Harman's one factor test.

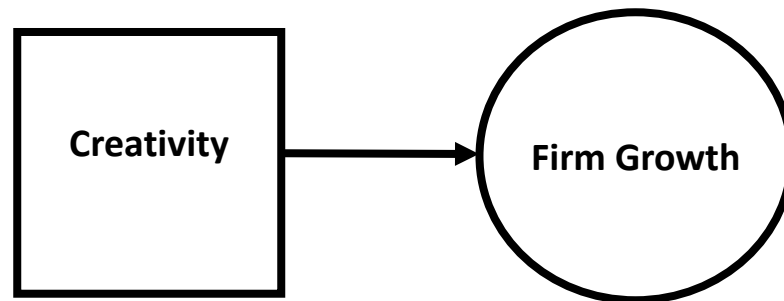
Endogeneity

- * **What is it?**
- * **Why does it hurt our thesis?**
- * **How do we deal with it?**

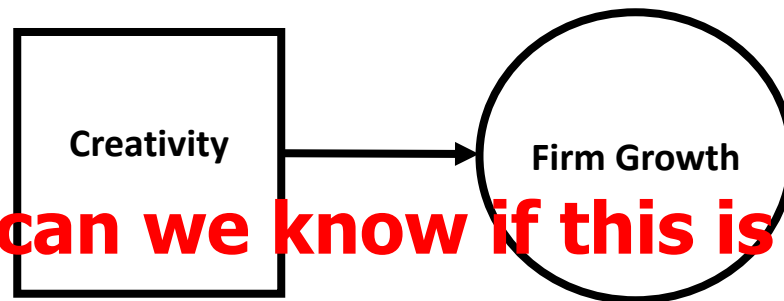


Endogeneity

Scenario 1: Omitted variables



Endogeneity



How can we know if this is correct?

Endogeneity

Scenario 1: omitted variable bias



Endogeneity



Endogeneity



Endogeneity



Endogeneity

Data

50 observations

Disc whole

Sound not heard – 19 times

Sound heard - 2 times



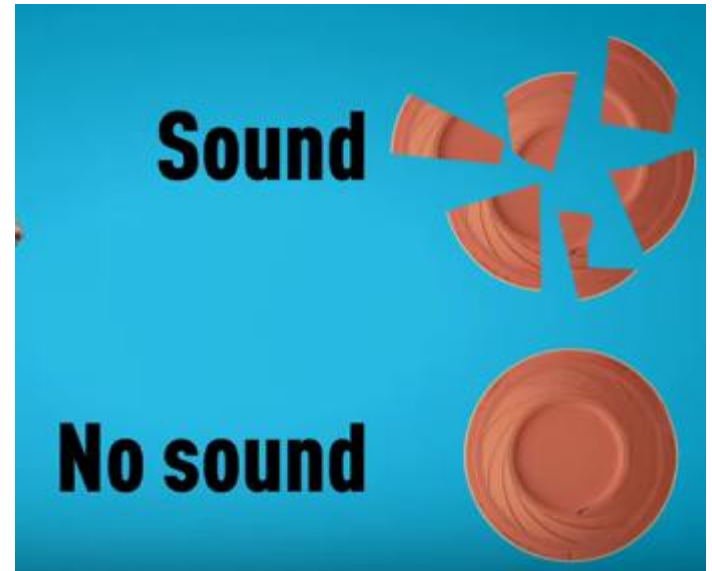
Disc shattered

Sound heard 29

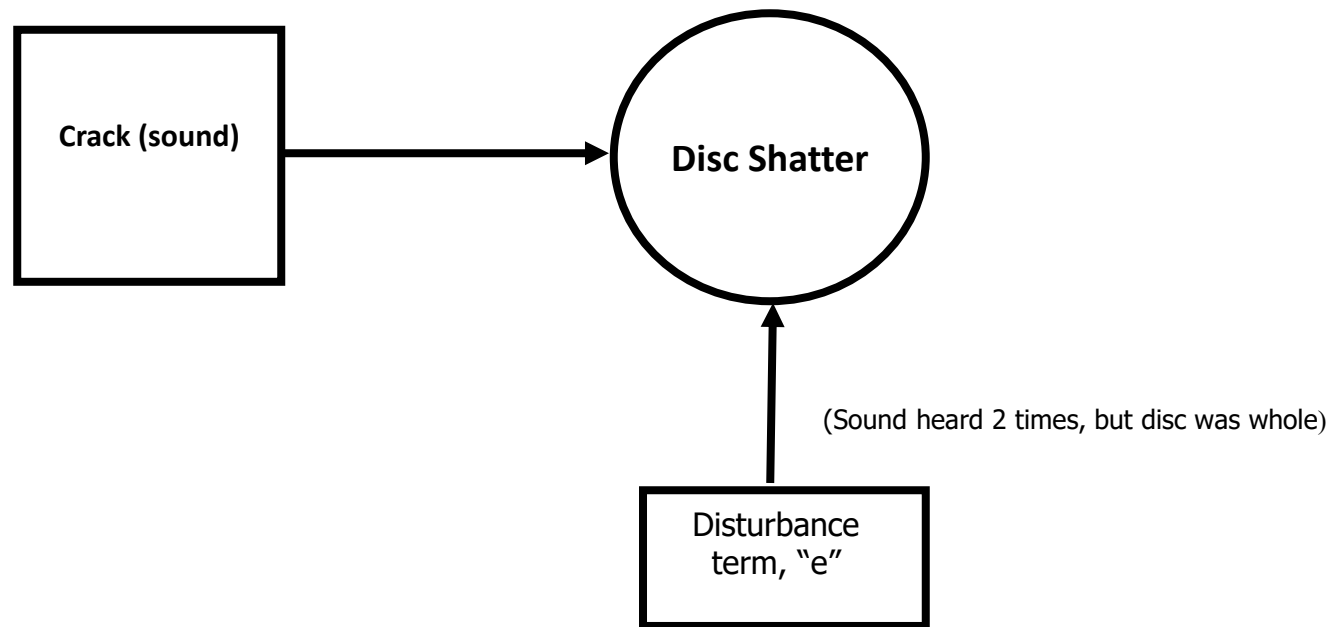


Endogeneity

- * Can you conclude that the sound makes the disc shatter based on your observations?
- * The observed correlation is very strong (phi coefficient 0.92)
- * But does it reflect the true relationship?



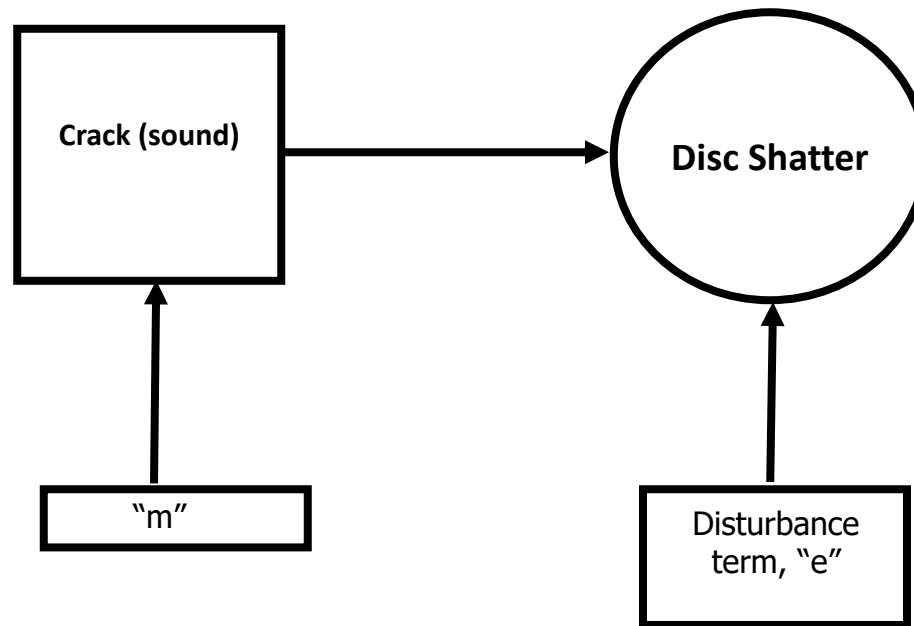
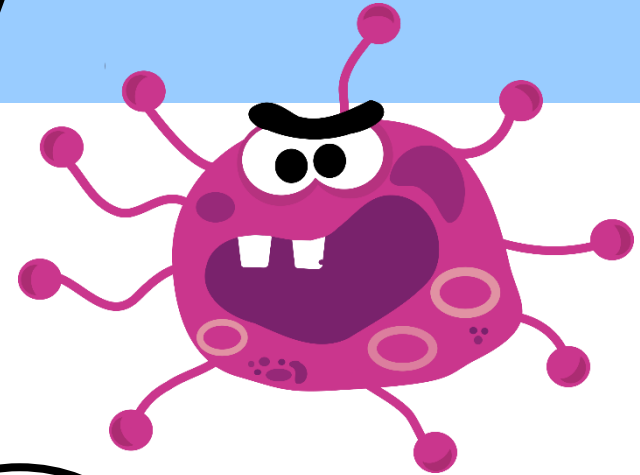
Endogeneity



Endogeneity

Problem:

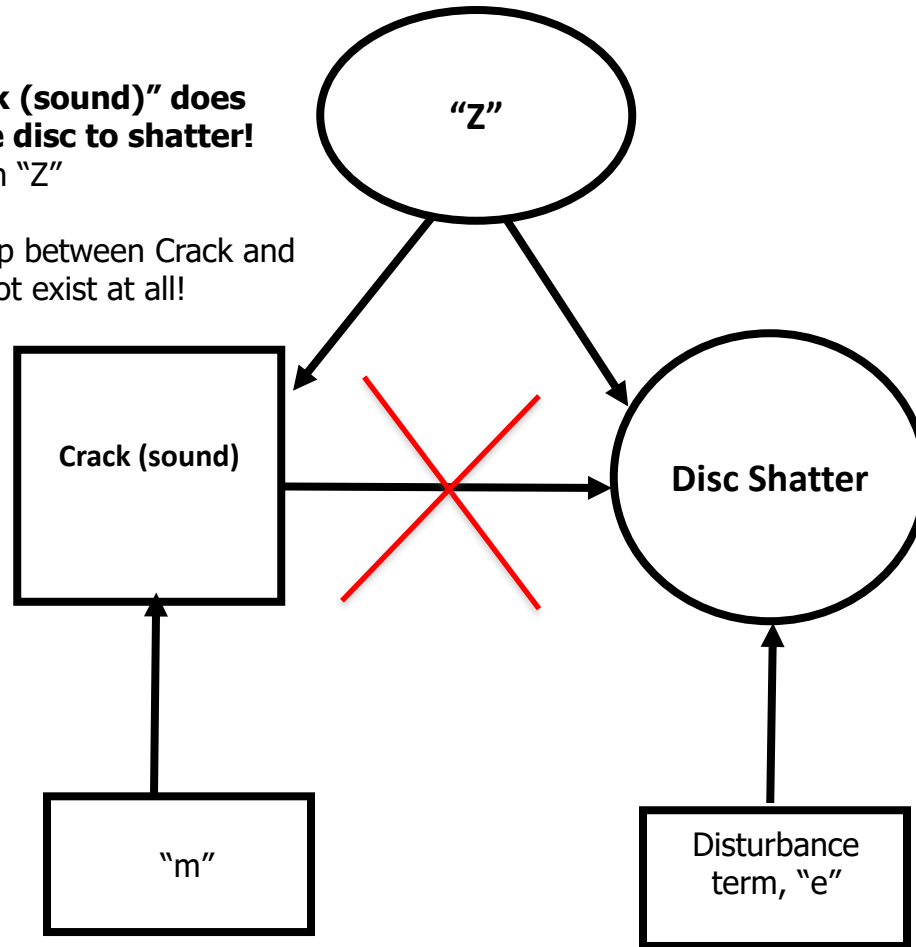
Crack is not exogenous (an independent variable), i.e. it depends on something.



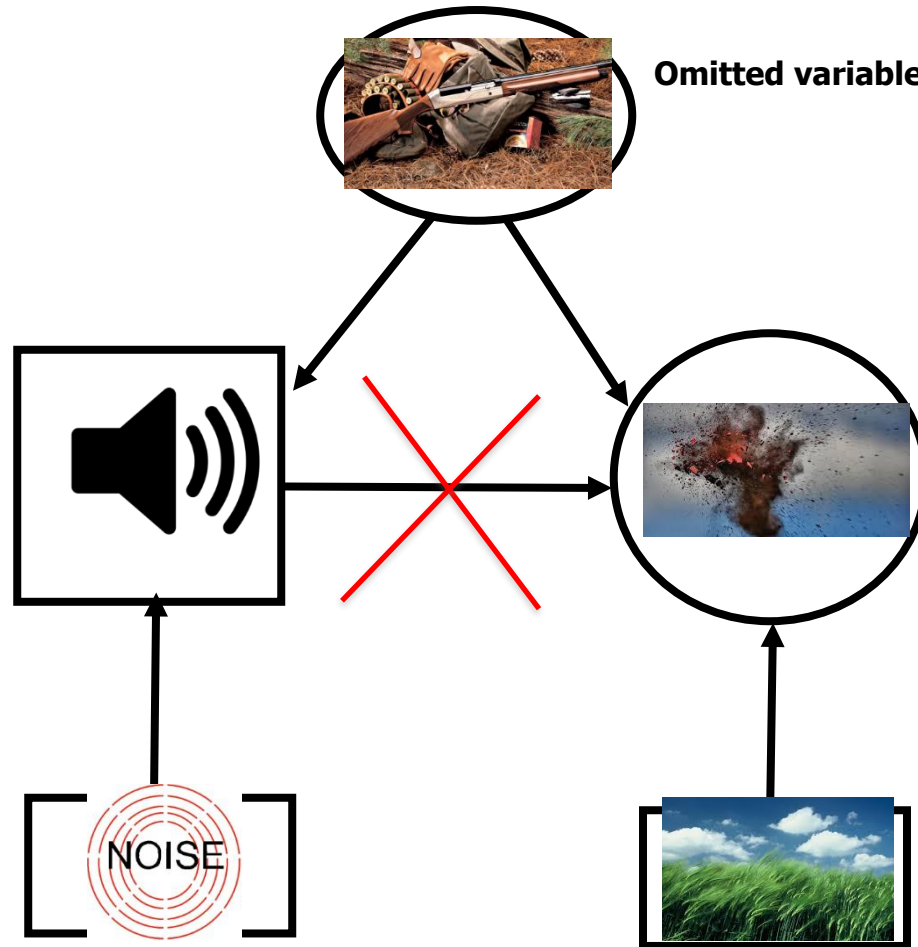
Endogeneity

In fact "Crack (sound)" does not cause the disc to shatter!
It goes through "Z"

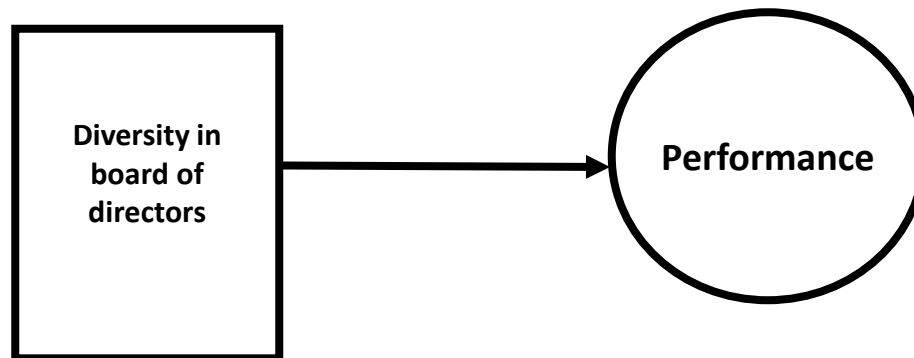
The relationship between Crack and Shatter does not exist at all!



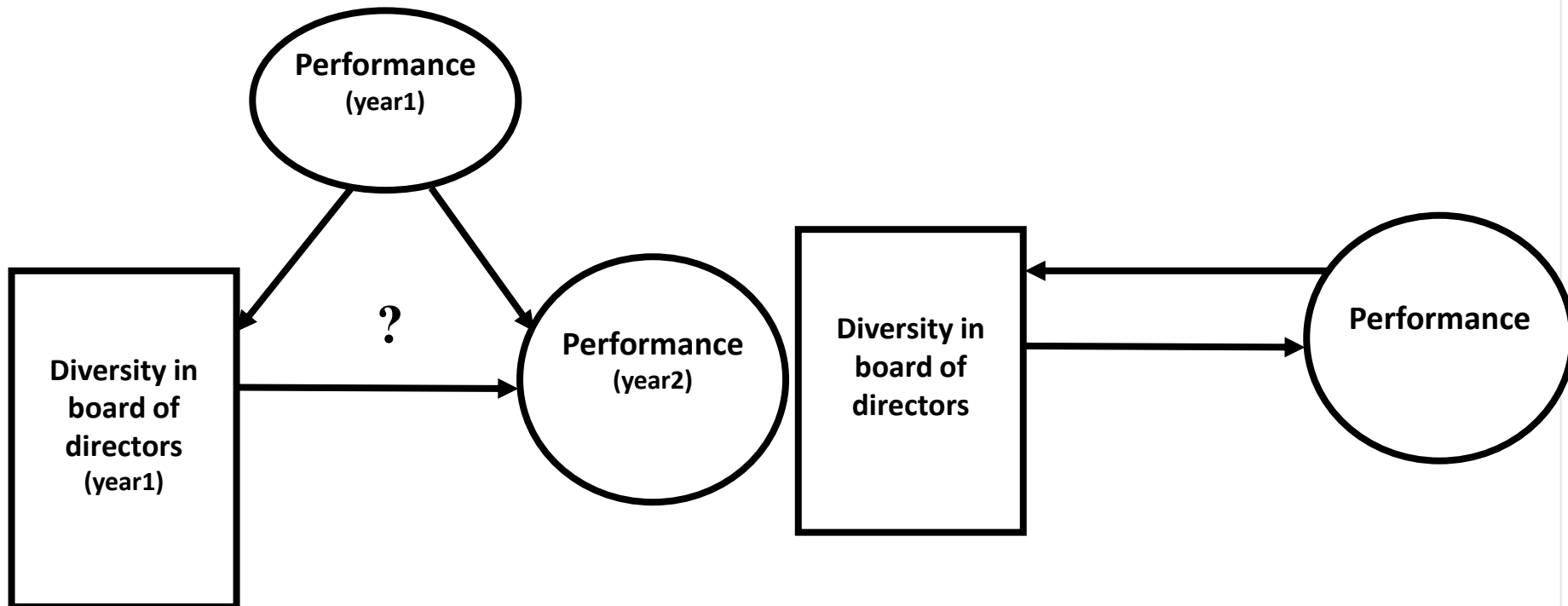
Endogeneity



Endogeneity

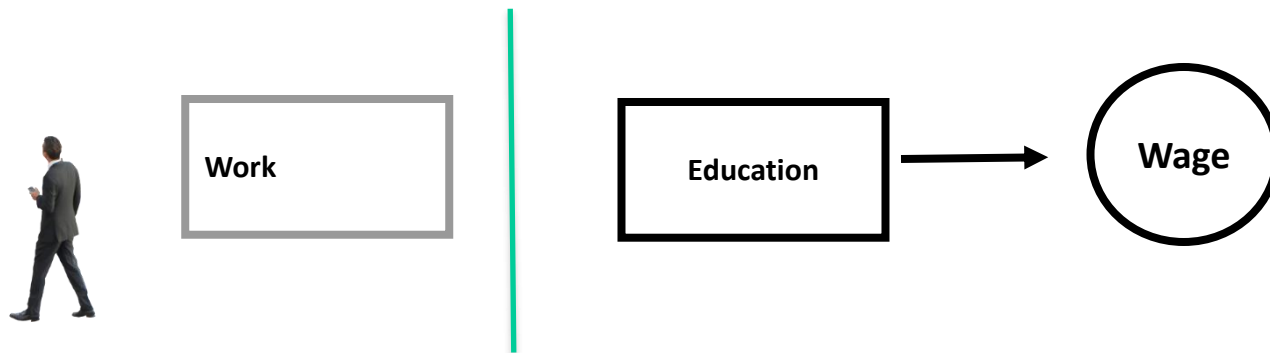


Endogeneity



Endogeneity

Scenario 2: Selection bias



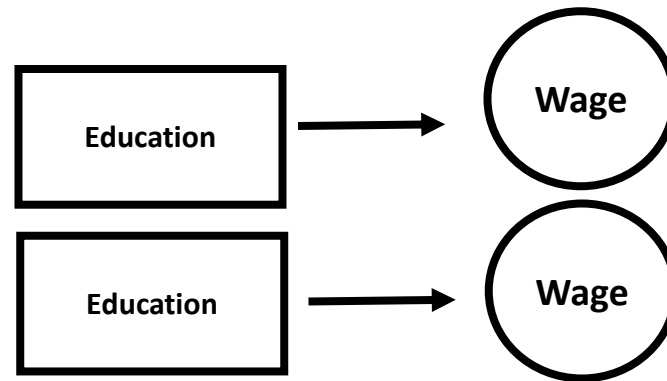
Endogeneity

Scenario 2: Selection bias



Work

No work



What would these individuals in the “no work” group had in wage, had they chosen to work?

Selection bias example: liability of foreignness

Empirical study designed to assess the Liability of Foreignness

Based on the idea that multinational firms, when they enter foreign markets, face additional costs over those incurred by domestic incumbents (Hymer, 1976; Zaheer, 1995; Mezias, 2002)

Uncertainty: Managerial inability to acquire mindsets of culturally distant customers, partners, and suppliers, and to understand the local 'rules of the game'

Discrimination: Consumer and workforce discrimination against firms and products from distant cultures, policies that discriminate against foreign firms

Complexity: Organizational costs related to employees' cultural differences (e.g., lack of trust, agency costs), to travel and coordination costs inflated by geographic distance, and to lack of face-to-face contact

**Zaheer,
1995:**

Hypothesis 1: Firms trading
rooms will be less profitable

this line of business. I also controlled for trading room size. A one-way analysis of variance comparing profits per trader in the foreign and local rooms, with the age and the size of the trading room as covariates, yielded a cell mean of 13.9 for foreign trading rooms and 14.42 for local rooms ($F = 7.5, p < .05$). These results support Hypothesis 1, which predicts that trading rooms operating overseas will be less profitable than rooms operating in their home country.

Going abroad is a choice! What about the firms that could have gone abroad but choose not to? Did they anticipate the liability of foreignness?

Gender pay-gap

Women are said to earn around 80% of what their male counterparts earn.

Do you believe you will earn less than the males in this room?



Could there be choices that women make, on average, that men do not that explain part of the estimated gap?

Endogeneity

What to do? Problem of the unknown! Be practical but be aware of these issues.

- 1. Experiment settings (best alternative)**
- 2. Econometric approaches (e.g. fixed effects, instrumental variable approach)**



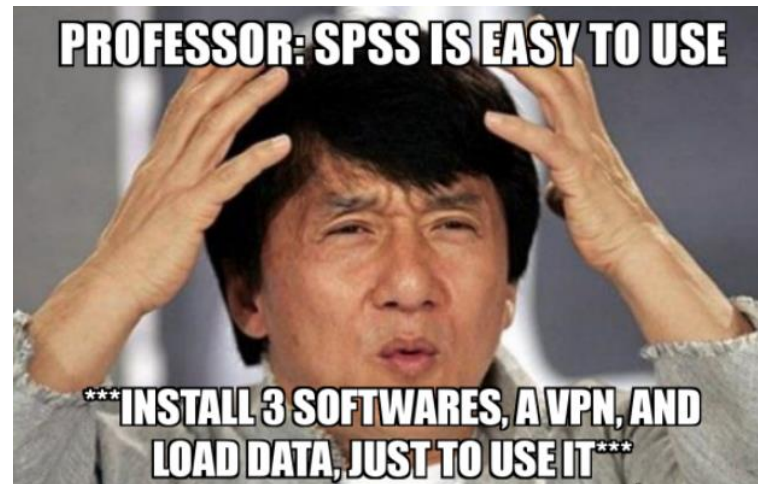


Part II

A primer to SPSS

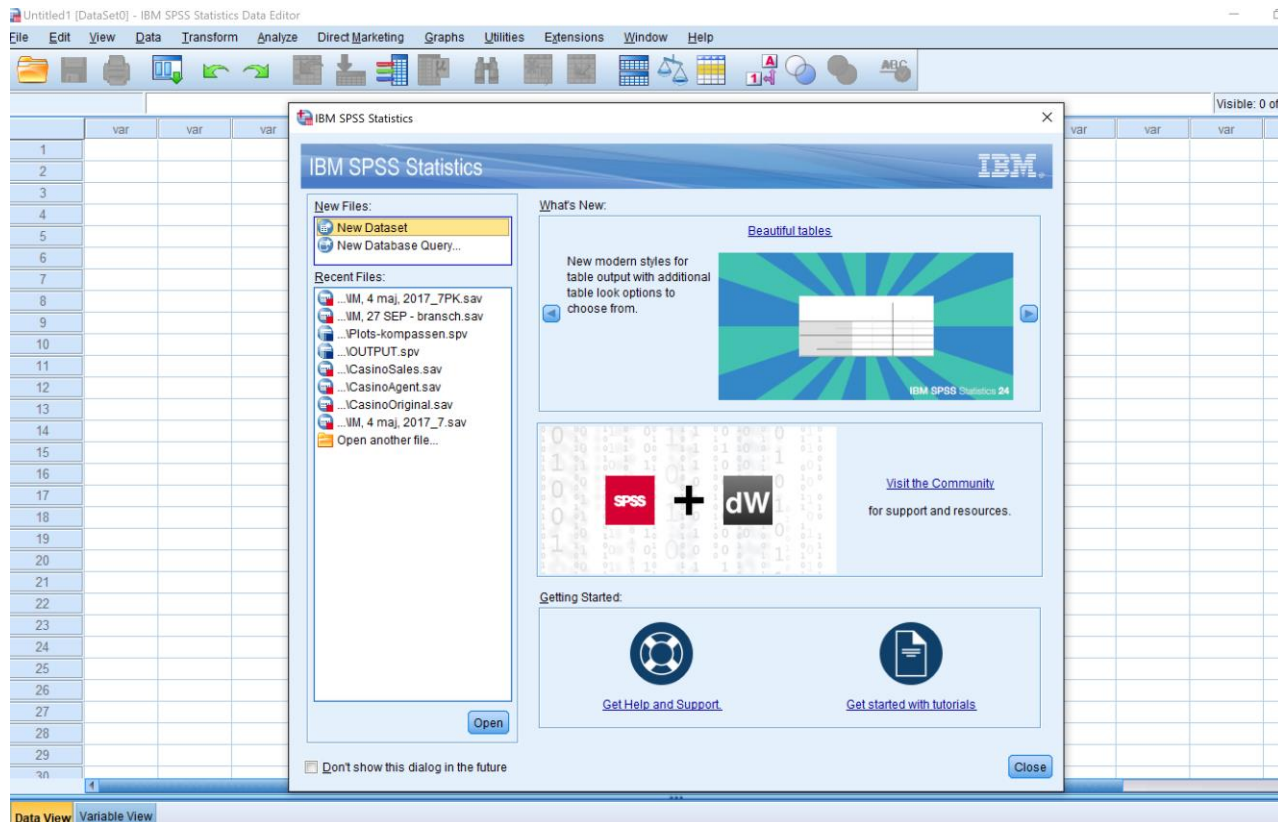
Getting to know your data

Instructions on
how to access
SPSS found on
Canvas...



Getting to know your data

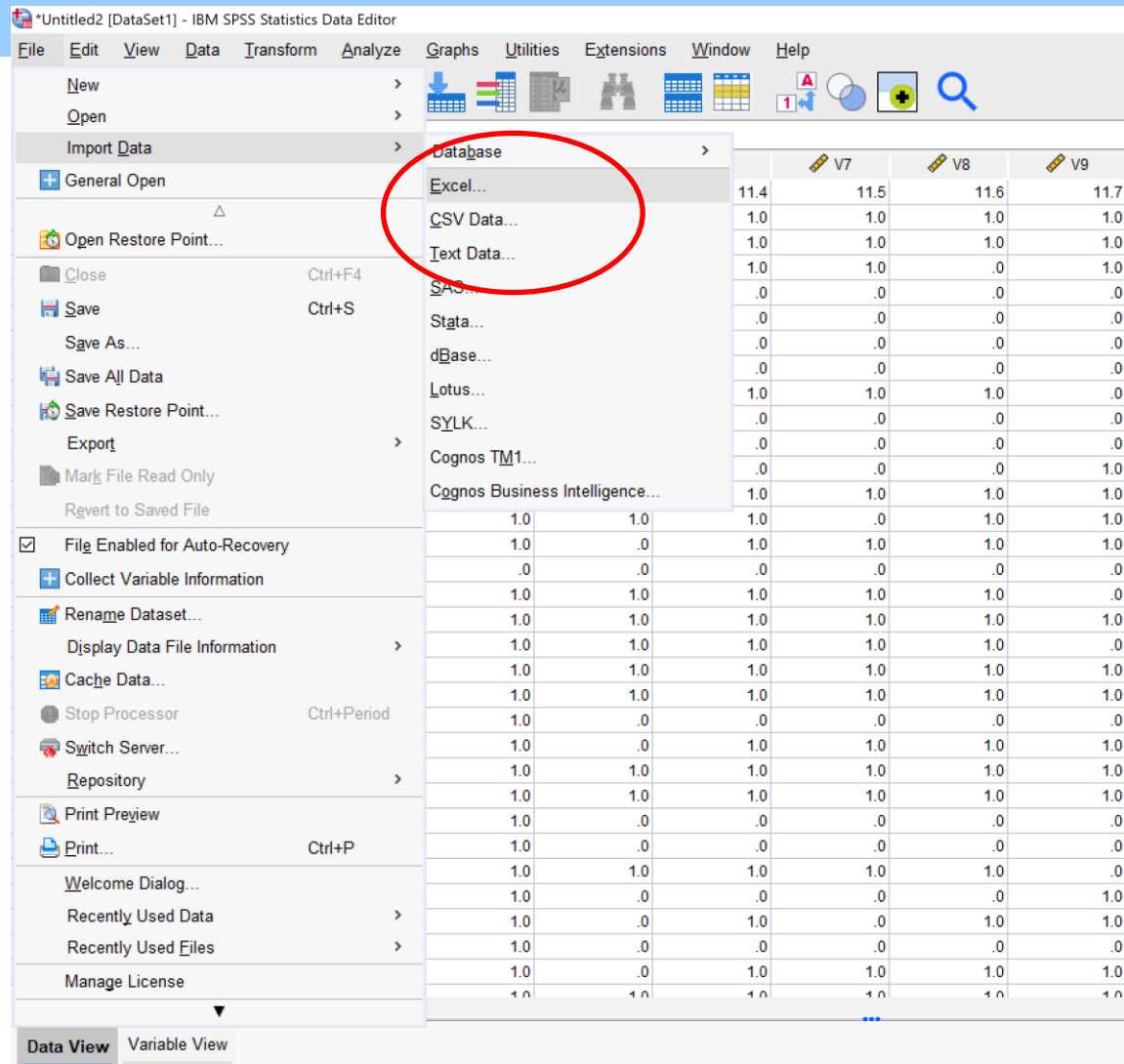
Open new dataset or a saved one from the loading screen



Getting to know your data

Import data

File → Import Data → Excel



Getting to know your data

The 3 main documents you work with

Survey

194	1	180	11	2	2	3	1	3
195	45	1881	12	2	2	3	3	3
196	1	180	11	3	2	2	1	3
197	1	180	11	2	2	3	7	3
198	20	840	21	4	2	3	1	1
199	39	1497	23	2	2	2	1	3
200	11	642	21	2	2	2	7	3
201	33	1480	23	2	2	1	2	3
202	4	380	12	2	2	3	1	3
203	59	2281	32	2	2	3	2	2
204	52	2081	31	2	2	3	2	3
205	33	1480	23	2	2	3	7	3
206	1	180	11	2	2	3	1	1
207	20	881	21	3	2	2	1	3
208	64	2460	33	2	2	2	1	3
209	2	188	11	2	2	2	1	3
210	1	184	11	2	2	3	2	2
211	58	2184	31	2	2	3	1	3
212	56	2180	31	3	2	2	1	3
213	7	484	12	3	2	2	1	3
214	38	1498	23	2	2	2	5	3
215	33	1419	23	2	2	2	3	3

Codebook

Question	Variable Number	Code Description	Variable Name
	1	Record number	RECNUM
	2	Respondent number	RESID
1	3	5 digit zip code 99999 = Missing	ZIP
2	4	2 digit birth year 99 = Missing	BIRTH
3	5	Gender 1 = Male 2 = Female 9 = Missing	GENDER
4	6	Marital status 1 = Married 2 = Widow(er) 3 = Divorced 4 = Separated 5 = Never married 9 = Missing	MARITAL
5	7	Own-Rent 1 = Own 2 = Rent 3 = Provided 9 = Missing	HOUSING

SPSS

Untitled2 [DataSet1] - IBM SPSS Statistics Data Editor

File Edit View Data Transform Analyze Direct Marketing

	VAR_ID	VAR_1	VAR_UNI T	VAR_4
1	1,00	1,00	1,00 Swe	
2	2,00	1,00	2,00 Swe	
3	3,00	1,00	3,00 Swe	
4	4,00	1,00	4,00 Swe	
5	5,00	1,00	5,00 Swe	
6	6,00	1,00	6,00 Swe	
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Data View Variable View

Data view

*Untitled2 [DataSet1] - IBM SPSS Statistics Data Editor

	VAR_ID	VAR_1	VAR_UNI T	VAR_4
1	1,00	1,00	1,00	Swe
2	2,00	1,00	2,00	Swe
3	3,00	1,00	3,00	Swe
4	4,00	1,00	4,00	Swe
5	5,00	1,00	5,00	Swe
6	6,00	1,00	6,00	Swe
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Data View Variable View

Rows →

Columns ↑

Clicking on
Variable View
will take you to...

*Untitled2 [DataSet1] - IBM SPSS Statistics Data Editor

File Edit View Data Transform Analyze Direct Marketing G

Folder Save Print New Undo Redo Paste Sort Filter Split View

	VAR_ID	VAR_1	VAR_UNI T	VAR_4
1	1,00	1,00	1,00	Swe
2	2,00	1,00	2,00	Swe
3	3,00	1,00	3,00	Swe
4	4,00	1,00	4,00	Swe
5	5,00	1,00	5,00	Swe
6	6,00	1,00	6,00	Swe
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Data View Variable View

Getting to know your data

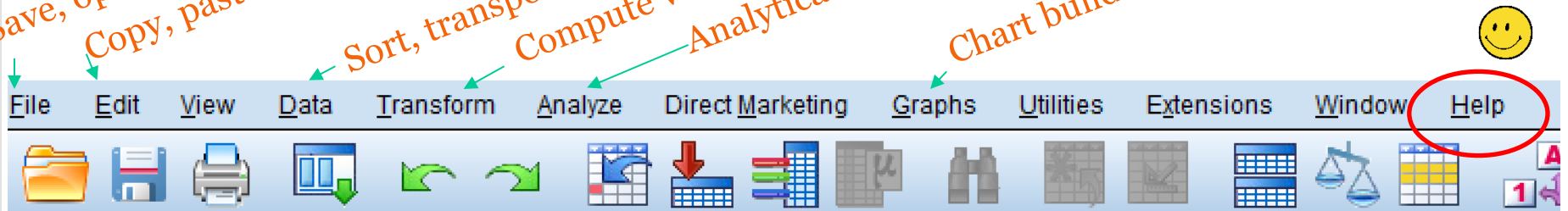
Variable view

File Edit View Data Transform Analyze Direct Marketing Graphs Utilities Extensions Window Help											
                   											
	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
1	VAR_ID	Numeric	8	2	Identifier	None	-9,00	8	Right	Unknown	Input
2	VAR_1	Numeric	8	2	MNE	None	-9,00	8	Right	Unknown	Input
3	VAR_UNIT	Numeric	8	2	MNE unit number	None	-9,00	8	Right	Unknown	Input
4	VAR_4	String	3	0	Country	None	-9	10	Left	Nominal	Input

Getting to know your data

The tool bar (in both Variable View and Data View)

Save, open data
Copy, paste, find values, replace values
Sort, transpose, select responses, filter responses
Compute variables, recode variables
Analytical techniques – exploration of data, data testing
Chart builder



	VAR_ID	VAR_1	VAR_UNI T	VAR_4	var	var	var	var
1	1,00	1,00	1,00	Swe				
2	2,00	1,00	2,00	Swe				
3	3,00	1,00	3,00	Swe				
4	4,00	1,00	4,00	Swe				

Getting to know your data

Detecting errors - respondent

	Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align
1	record	Numeric	7	0	Record number	None	None	8	Right
2	age	Numeric	19	0	Hur gammal är du?	None	None	8	Right
3	gender	Numeric	1	0	Är du man eller kvinna?	{1, Man}...	None	8	Right
4	zipcode	Numeric	19	0	Vilket post				Right
5	age_group	Numeric	1	0	Hur gammal				Right
6	SE_County	Numeric	3	0	Län:				Right
7	SE_aregion	Numeric	2	0	Aregion:				Right
8	SE_municip...	Numeric	4	0	Kommun				Right
9	SE_nuts2	Numeric	2	0	Riksområd				Right
10	SE_occupat...	Numeric	2	0	Vilken sys				Right
11	work_sector	Numeric	2	0	Arbetar du				Right
12	Q3	Numeric	1	0	Vilken är d				Right
13	Q4	Numeric	1	0	Hur länge h				Right
14	Q5	Numeric	1	0	Vilken är d				Right
15	Q6	Numeric	2	0	Vilket är d				Right
16	Q6r10oe	String	1536	0	Vilket är d				Right
17	Q7	Numeric	1	0	Företagsst				Right
18	Q8	Numeric	2	0	Vilken yrke				Right
19	Q8r22oe	String	1536	0	Vilken yrke				Right
20	A1xA7r1	Numeric	1	0	Vårt företa				Right
21	A1xA7r2	Numeric	1	0	Vårt företa				Right
22	A1xA7r3	Numeric	1	0	I vårt företa				Right
23	A1xA7r4	Numeric	1	0	Min chef g				Right
24	A1xA7r5	Numeric	1	0	Min chef berättar för sin chef om min arbetsprestation är...	{1, 1 = instä...	9	8	Right
25	A1xA7r6	Numeric	1	0	Jag får prestationsbaserad belöning på mitt arbete - Ang...	{1, 1 = instä...	9	8	Right
26	A1xA7r7	Numeric	1	0	I vårt företag finns det ofta gott om pengar för att utveck...	{1, 1 = instä...	9	8	Right
27	A8	Numeric	1	0	I vårt företag har anställda formellt avsatt tid för att arbet...	{1, Ja}...	9	8	Right
28	A9	Numeric	1	0	I vårt företag finns det en separat enhet vars huvuduppgif...	{1, Ja}...	9	8	Right

Value Labels

Value: 1

Label: Sälj/Marknad

1 = "Sälj/Marknad"

2 = "Företagsledning"

3 = "Tillverkning"

4 = "Forskning/utveckling"

5 = "HR"

6 = "IT"

7 = "Inköp/Logistik"

8 = "Finans/Juridik"

9 = "Redovisning/Admin"

10 = "Annat"

Add

Change

Remove

Spelling...

OK Cancel Help

Getting to know your data

Detecting errors - respondent

	Function								Function, if something else	industry	
194	1	180	11	2	2	3	1	3	1	4	10
195	45	1881	12	2	2	3	3	3	10 Utbildning	4	16
196	1	180	11	3	2	2	1	3	10 Make up	4	22 Make up
197	1	180	11	2	2	3	7	3	4	4	3
198	20	840	21	4	2	3	1	1	6	2	9
199	39	1497	23	2	2	2	1	3	3	4	3
200	11	642	21	2	2	2	7	3	3	4	3
201	33	1480	23	2	2	1	2	3	10 Städning	2	16
202	4	380	12	2	2	3	1	3	6	4	8
203	59	2281	32	2	2	3	2	2	6	3	14
204	52	2081	31	2	2	3	2	3	6	4	22 IT
205	33	1480	23	2	2	3	7	3	9	3	22 Intresseorganisation
206	1	180	11	2	2	3	1	1	7	4	22 Personaltransport
207	20	881	21	3	2	2	1	3	10 Bajs	2	22 Bajspackare
208	64	2460	33	2	2	2	1	3	4	4	3
209	2	188	11	2	2	2	1	3	10	2	17
210	1	184	11	2	2	3	2	2	9	4	7
211	58	2184	31	2	2	3	1	3	10 Kundsupport	3	20
212	56	2180	31	3	2	2	1	3	10 Assistans	4	20
213	7	484	12	3	2	2	1	3	1	4	20
214	38	1498	23	2	2	2	5	3	3	3	3
215	33	1419	23	2	2	2	3	3	10 Service	3	20

Getting to know your data

Detecting errors – researcher

Questionnaire

Codebook

Data

Scale G

Listed below are a number of statements concerning personal attitudes and traits. Read each item and decide whether the statement is true or false as it relates to you.

Circle either True (T) or False (F) for each statement

- | | | | |
|---|---|-----|---|
| T | F | 1. | I'm always willing to admit it when I make a mistake |
| T | F | 2. | I always try to practice what I preach |
| T | F | 3. | I never resent being asked to return a favour |
| T | F | 4. | I have never been irked when people expressed ideas very different from |
| T | F | 5. | I have never deliberately said something that hurt someone's feelings |
| T | F | 6. | I like to gossip at times |
| T | F | 7. | There have been occasions when I took advantage of someone |
| T | F | 8. | I sometimes try to get even rather than forgive and forget |
| T | F | 9. | At times I have really insisted on having things my own way |
| T | F | 10. | There have been occasions when I felt like smashing things |

Marlowe-Crowne Social
Desirability Scale

m1 to m10

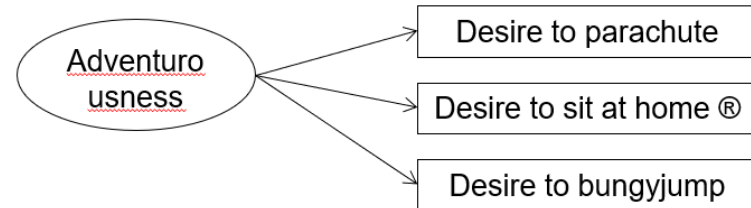
1=true, 2=false

m1–m10 only
contain the
values 0 and 1

Variable creation

Grouping of question is done with respect to **correlations**. Questions that may be considered to **measure the 'same' thing** should correlate to a greater extent with each other compared to questions measuring different things.

What it looks like conceptually



What it looks like in a questionnaire

Variable X

Scale D

Below are five statements with which you may agree or disagree. Using a 1 to 7 scale, indicate your agreement with each item by placing the appropriate number on the line next to that item.

strongly disagree 1 2 3 4 5 6 7 strongly agree

1. _____ In most ways my life is close to ideal
2. _____ The conditions of my life are excellent
3. _____ I am satisfied with my life
4. _____ So far I have got the important things I want in life
5. _____ If I could live my life again, I would change almost nothing

The easy solution

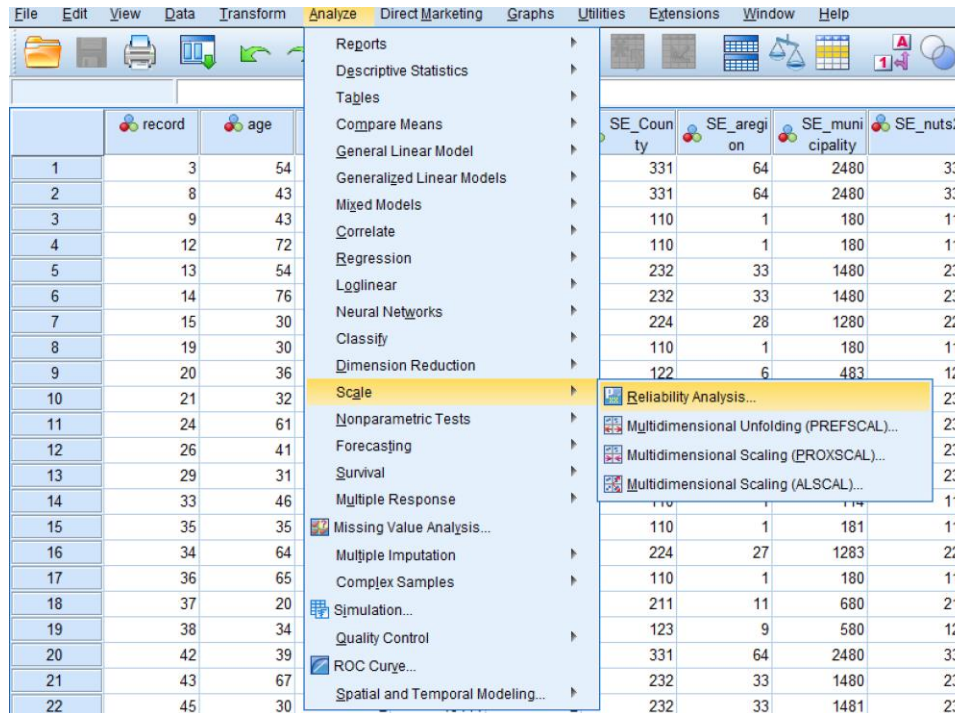
- Create a variable **using the mean or sum** of those questions for each respondent.
- You now have a **multifaceted variable** represented by a number of questions.
- You make **no claim on weights of questions**, discriminant validity between constructs or **convergent validity**.
- You can check **internal reliability**, and **you should for assignment 3**.



Internal reliability

Looks at how closely related a set of items are as a group - inter-item correlation

- Rule of thumb – above 0.7 (you get a value between 0-1 when running the check)
- Number of items influence the Cronbach alpha value (the more, the higher expected value).



	record	age
1	3	54
2	8	43
3	9	43
4	12	72
5	13	54
6	14	76
7	15	30
8	19	30
9	20	36
10	21	32
11	24	61
12	26	41
13	29	31
14	33	46
15	35	35
16	34	64
17	36	65
18	37	20
19	38	34
20	42	39
21	43	67
22	45	30

	SE_Coun ty	SE_aregi on	SE_muni cipality	SE_nuts2
	331	64	2480	33
	331	64	2480	33
	110	1	180	11
	110	1	180	11
	232	33	1480	23
	232	33	1480	23
	224	28	1280	22
	110	1	180	11
	122	6	483	12
	110	1	181	11
	224	27	1283	22
	110	1	180	11
	211	11	680	21
	123	9	580	12
	331	64	2480	33
	232	33	1480	23
	232	33	1481	23

Remember!

Look at how the Cronbach alpha would change if you would delete one question from the construct.

This will help you get the best fitting variable.

Internal reliability

SPSS Statistics Data Editor

Name	Type	Width	Decimals	Label	Values	Missing	Columns	Align	Measure	Role
Question1	Numeric	8	2		None	None	8	Right	Nominal	Input
Question2	Numeric	8	2		None	None	8	Right	Nominal	Input
Question3	Numeric	8	2		None	None	8	Right	Nominal	Input
NewVariable	Numeric	8	2							

Reliability Analysis

Items:

- Question1
- Question2
- Question3
- NewVariable

Statistics...

Model: Alpha

Scale label:

OK Paste Reset Cancel Help

Reliability Analysis: Statistics

Descriptives for

- ☐ Item
- ☐ Scale
- ☒ Scale if item deleted

Summaries

- ☐ Means
- ☐ Variances
- ☐ Covariances
- ☐ Correlations

ANOVA Table

- ☒ None
- ☐ E test
- ☐ Friedman chi-square
- ☐ Cochran chi-square

Interrater Agreement: Fleiss' Kappa

- ☐ Display agreement on individual categories
- ☐ Ignore string cases
- ☒ String category labels are displayed in uppercase

Asymptotic significance level (%): 95

Missing

- ☒ Exclude both user-missing and system missing values
- ☐ User-missing values are treated as valid

Model: Two-Way Mixed

Scale label:

- ☐ Hotelling's T-square
- ☐ Intra-class correlation coefficient
- ☐ Tukey's test of additivity

Type: Consistency

Confidence interval: 95 % Test value: 0

Continue Cancel Help



Internal reliability

In research...

tion). An extracted factor score was used to form the dependent variable. The internal reliability of the construct had a coefficient alpha of

Journal of International Business Studies

0.697, which can be considered satisfactory, given the recommendation of 0.7 as made by Nunnally (1978). The dependent variable builds on and extends the attention-based perspective on MNE headquarters (Bouquet & Birkinshaw, 2008; Bouquet et al., 2009; Ocasio, 1997). As discussed by

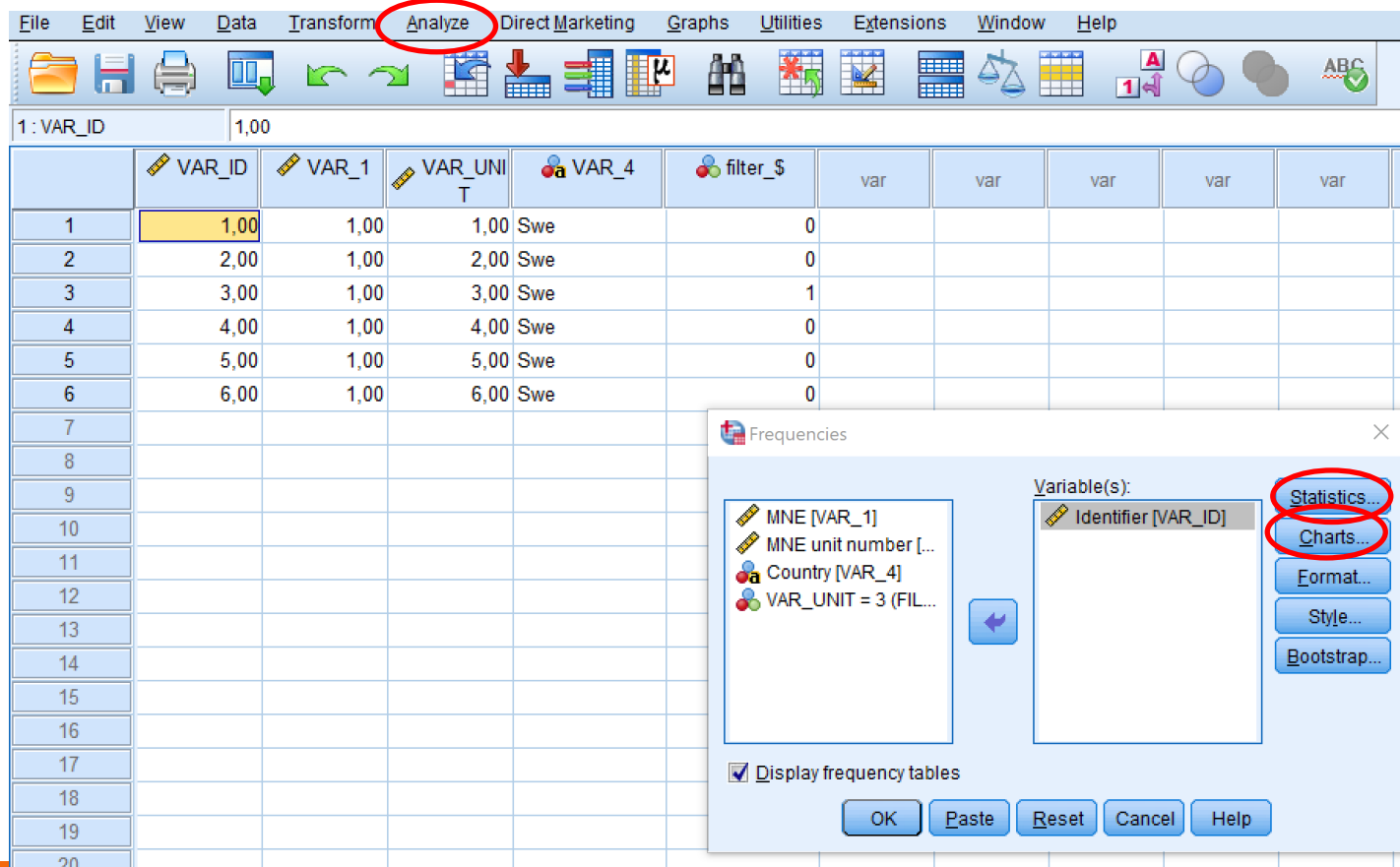
Cronbach alpha



Next step: data exploration

How to: Explore data by looking at Frequencies

Analyze → Descriptive statistics → Frequencies



File Edit View Data Transform **Analyze** Direct Marketing Graphs Utilities Extensions Window Help

1: VAR_ID 1,00

	VAR_ID	VAR_1	VAR_UNIT	VAR_4	filter_\$	var	var	var	var	var
1	1,00	1,00	1,00	Swe	0					
2	2,00	1,00	2,00	Swe	0					
3	3,00	1,00	3,00	Swe	1					
4	4,00	1,00	4,00	Swe	0					
5	5,00	1,00	5,00	Swe	0					
6	6,00	1,00	6,00	Swe	0					
7										
8										
9										
10										
11										
12										
13										
14										
15										
16										
17										
18										
19										
20										

Frequencies

Variable(s):

- MNE [VAR_1]
- MNE unit number [...]
- Country [VAR_4]
- VAR_UNIT = 3 (FIL...

Identifier [VAR_ID]

☒ Display frequency tables

OK Paste Reset Cancel Help

Statistics... Charts... Format... Style... Bootstrap...

Data exploration

Frequencies output

Depends on the options specified
but usually something like this →

→ Frequencies

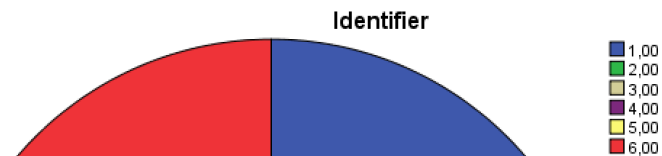
Statistics

Identifier		
N	Valid	6
	Missing	0
Mean		3,5000
Median		3,5000
Mode		1,00 ^a
Minimum		1,00
Maximum		6,00

a. Multiple modes exist.
The smallest value
is shown.

Identifier

		Frequency	Percent	Valid Percent	Cumulative Percent
Valid	1,00	1	16,7	16,7	16,7
	2,00	1	16,7	16,7	33,3
	3,00	1	16,7	16,7	50,0
	4,00	1	16,7	16,7	66,7
	5,00	1	16,7	16,7	83,3
	6,00	1	16,7	16,7	100,0
Total		6	100,0	100,0	



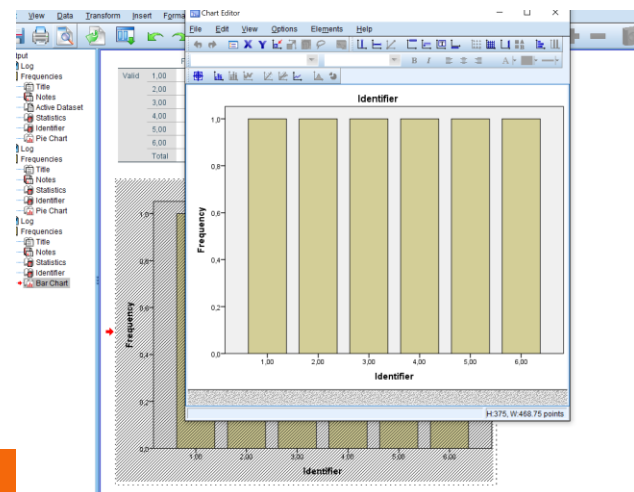
Data exploration

Bar charts, Pie charts and Histograms

From the “Frequencies” option it is possible to attain Bar charts, Histograms and Pie charts.

Double click the generated chart to open the “Chart Editor”

Change fonts, placing, colors and more to make your chart look professional.

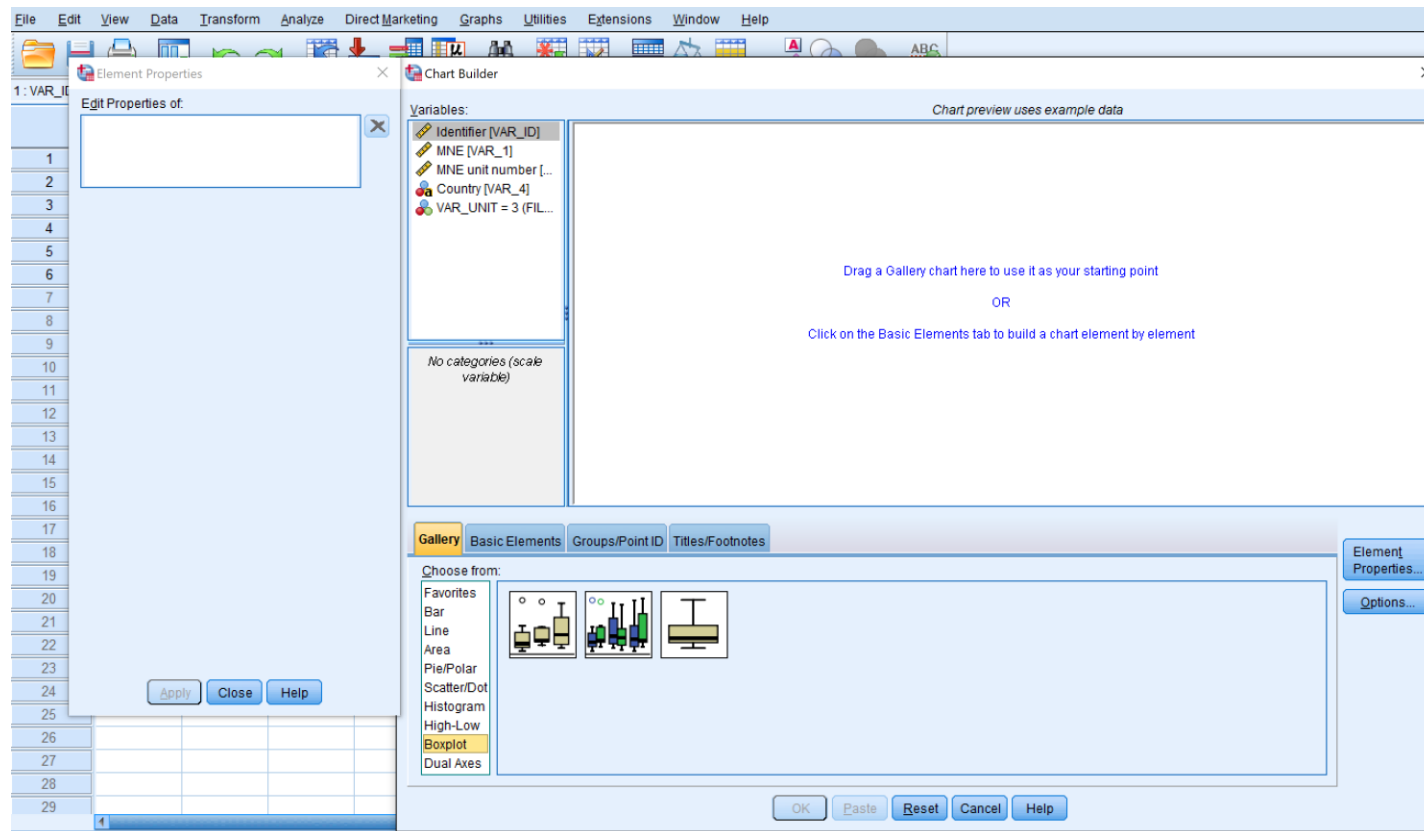


Data exploration

Boxplots

Graphs → Legacy dialogs → Boxplot

Or Graphs → Chart Builder → (find boxplot in lower left corner)



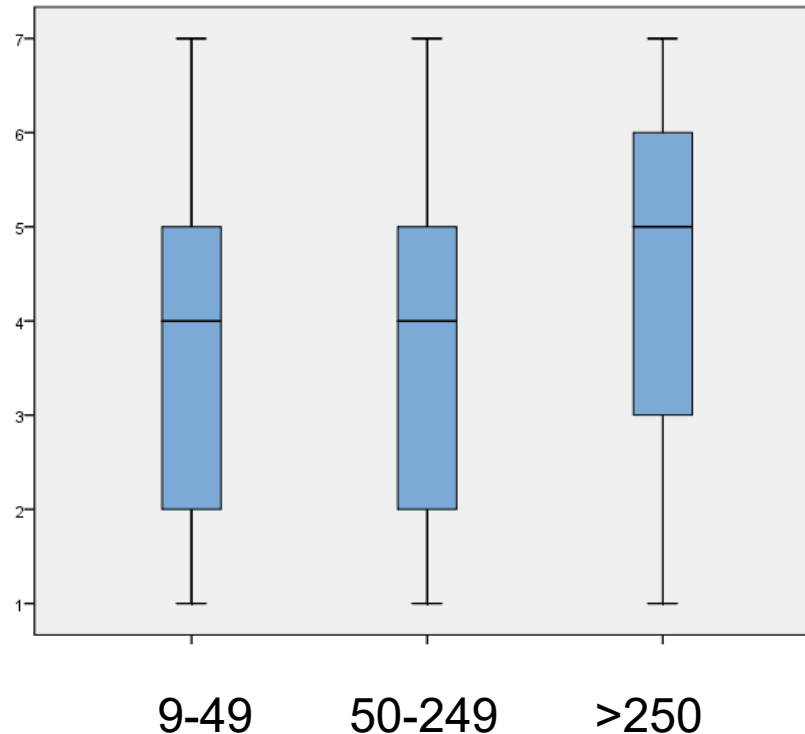
Data exploration

Boxplots

Scale: 1-7

Question:

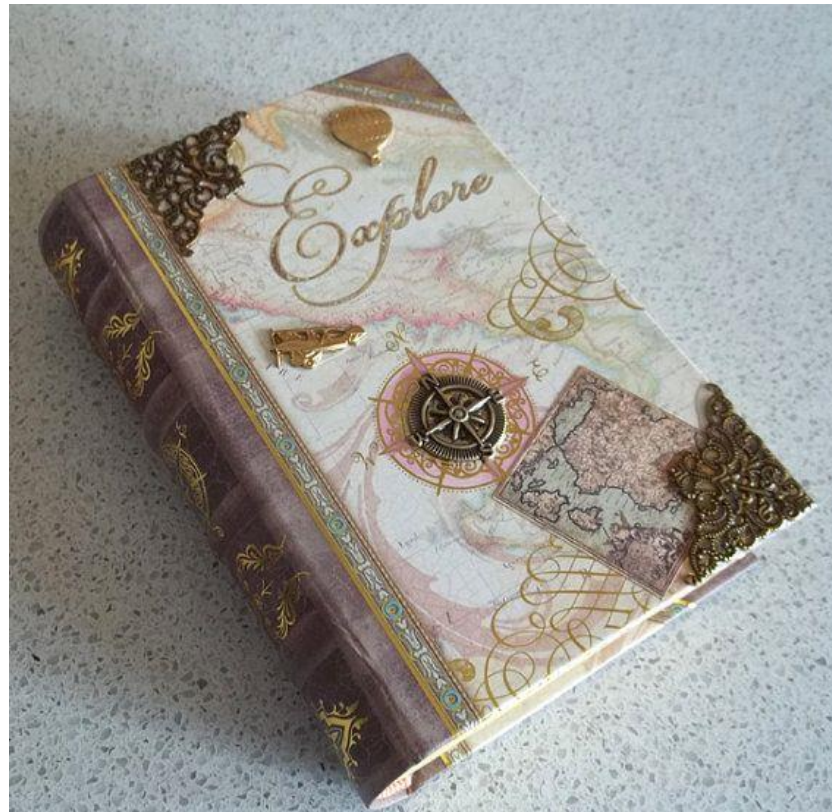
In our firm we emphasize the recruitment of staff with competencies in developing new products and services



Question: Firm size

(Employees)

Reporting our Exploration



Descriptives for Information

In text

In order to assess the distribution of technological contributions across foreign subsidiaries of the MNC, we started by examining the complete US patenting records among all of the advanced foreign subsidiaries from a sample of the most R&D-intensive Swedish firms throughout the period between 1893 and 2008.⁵ The search extracted 21 firms and identified 368 foreign subsidiaries with registered patenting activity for the examined period (for further details about the sample, see [Appendix A](#)). These are all advanced foreign units in the sense that they have proven their ability to make significant contributions to the state of the art in various fields of technology (a prerequisite for patentability). Most of them can also be expected to display a comparatively high complexity of activities and technological work of competence-creating subsidiaries ([Cantwell and Mudambi, 2005, 2011](#)). The age of an advanced subsidiary, as measured from its first recorded US patent, ranged from one year to 100 years, with an average age of 19 years and a median of 13 years.

The foreign subsidiaries included a total of 237 greenfield subsidiaries and 131 subsidiaries that were added to the multinational groups through acquisitions. The first step of the analyses was



Superstar inventors—Towards a people-centric perspective on the geography of technological renewal in the multinational corporation^a

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^b Copenhagen Business School, Department of Strategic Management and Globalization, Farvej 14A, 2000 Frederiksberg, Denmark

The identification of inventive activity within the selected subsidiaries covered the period from when they started as technologically advanced units until 2008 (in one of the cases, Alfa Laval's US subsidiary, the first available data at the individual level was from 1920 and merger activity truncated all observations in 1993). Patenting activity at the level of individual inventors was identified by searching the Thomson Innovation database for patents associated with a variety of names connected to the sample firms and subsidiaries. This search generated lists comprising more than 13,500 patents, which were subsequently separated from those patents that were found to be unrelated to the sample firms.

A total of 4174 patents remained for further analysis, ranging from 106 in the case of Sandvik's German subsidiary to 2528 in Ericsson's US subsidiary. Each patent was analyzed with regard to its inventors,⁸ which ranged from one to 11 per patent. The order of inventors as they appeared on the patent was registered and co-inventors from locations other than those associated with the selected subsidiaries were recorded. In addition, each patent was compared to the complete patent portfolio of the MNC in order to distinguish which patents represented entries into classes of technology that were new to the entire multinational group (these entries into new technologies were the same as those used to derive the group of exceptionally inventive subsidiaries). This eventually enabled a formalized test that aimed to distinguish the role of individual inventors for the more significant broadening of the MNC's technological portfolio.

Descriptives for Information

In appendices

and how they can have a decisive influence on its technological, strategic, and organizational development. Perhaps the most significant contribution of this study is that it highlights the important role played by individuals and foreign superstar inventors in the development of the overall technological portfolio of the MNC.

Appendix A.

The sample of consolidated Swedish MNCs that provided the basis for the sampling of superstar subsidiaries consisted of the following firms:

Firm ^a	Principal field of industrial activity	Number of greenfield/acquired subsidiaries
AGA (1904) ^b	Industrial gases	7/2
Alfa Laval (1878)	Separators, agricultural equipment	13/8
ASEA (1883)	Power generation and distribution equipment	14/3
Astra (1913)	Pharmaceuticals	19/1
Atlas Copco (1873)	Pneumatic and hydraulic equipment	14/9
Electrolux (1910)	White goods, home appliances	25/14
Ericsson (1876)	Telecommunication equipment	25/6
ESAB (1904)	Welding equipment	3/7
Fagersta (1873)	Metals, rock drills	1/0
MoDo (1873)	Pulp and paper	4/1
Perstorp (1880)	Chemicals, conglomerate	8/3
Pharmacia (1911)	Pharmaceuticals	13/0
PLM (1919)	Packaging material	3/1
Saab-Scania (1891)	Automotive products, aircraft	2/1
Sandvik (1862)	Specialty steel and metals, hard materials	20/34
SCA (1925)	Pulp and paper	14/14
SKF (1905)	Ball and roller bearings	23/11
Stora (1888)	Pulp and paper	1/7
Tetra Pak (1946)	Liquid packaging machinery	10/0
Trelleborg (1905)	Rubber products, conglomerate	8/5
Volvo (1915)	Automotive products, food	10/4

^a Years in parentheses indicate the year of establishment.

^b AGA was acquired by Linde in 2000 and observations were truncated in that year. Other sample firms with truncated observations include ASEA (1988, merged with Swiss Brown Boveri et Cie.), Alfa Laval (1993, acquired by Tetra Pak), Astra (1999, merged with Zeneca Group), ESAB (1994, acquired by Charter), MoDo (2000, acquired by Metsä), Perstorp (2001, acquired by Sydsvenska Kemi), Pharmacia (1990, merged with Kabi Vitrum), PLM (1998, acquired by Rexam), Scania (1990, car division acquired by GM), Stora (1998, merged with Enso), Tetra Pak (1993, acquired Alfa Laval), and Volvo (1999, car division acquired by Ford).

Descriptives for Information

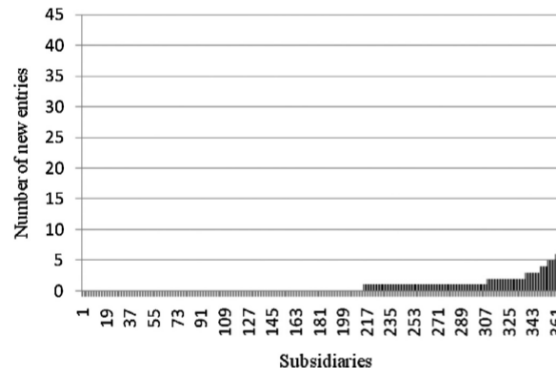
In tables and figures

Table 3
Extent of cross-regional and cross-country collaborations among inventors.

	Full sample	Superstar sample	Restricted sample ^a
Cross-national collaboration	0.089	0.16	0.064
Cross-regional collaboration	0.138	0.23	0.111

Values expressed in percentages of all patents.

^a Full sample excluding superstar inventors.



Bar chart (with
very tiny spaces
between bars)

Fig. 1. Number of entries into classes of technology new to the entire multinational group – all subsidiaries.

Descriptives as Results

- Use simple descriptives / overview tables of your data
- ***Empirical studies need not contain statistical analysis to enable researchers to draw valid inferences!***



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Research Policy 34 (2005) 395–410



Foreign direct investment in industrial research and development: A study of German MNCs
Björn Ambos*

Table 6

Co-location of R&D and production sites within firms

	Capability exploiting	Capability augmenting	Total
Production facility at site	77	26	103
Separate entity	14	13	27



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journal homepage: www.elsevier.com/locate/respol



Superstar inventors—Towards a people-centric perspective on the geography of technological renewal in the multinational corporation*

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^b Copenhagen Business School, Department of Strategic Management and Globalization, Kløvsgade 14A, 2000 Frederiksberg, Denmark

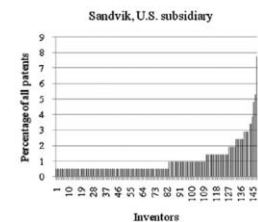
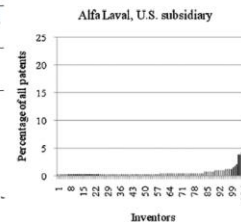


Table 3

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Cross-national collaboration	0.089	0.16	0.064
Cross-regional collaboration	0.138	0.23	0.111

Values expressed in percentages of all patents.

^a Full sample excluding superstar inventors.

Descriptives as Results



Table 2: Home Region distribution of sales of the world's TNC index, 2001

500 Rank	Corporation	Home Country	intra-regional 2001*	NA (%)	EUR (%)	AP (%)	C
1	147 Nokia	Finland	49.0	25.0 ¹	49.0	26.0	G
2	288 Roche	Switzerland	36.8	38.6	36.8	11.7	B
3	194 ABB	Switzerland	53.9	25.1	53.9	11.3	D
4	143 Philips Electronics	Netherlands	43.0	28.7 ^a	43.0	21.5	G
5	263 Nortel Networks	Canada	54.4	54.4 ^a	na	na	D
6	423 Stora Enso	Finland	69.2	19.5	69.2	7.1	D
7	301 AstraZeneca	UK	32.0	52.8 ^z	32.0	5.2 ^j	S
8	267 Volvo	Sweden	51.6	30.2	51.6	6.0	D
9	140 GlaxoSmithKline	UK	28.6	49.2 ^z	28.6	na	B
10	364 News Corp.	Australia	9.0	75.0 ^z	16.0 ^u	9.0	S
11	262 Diageo	UK	31.8	49.9	31.8	7.7	B
12	416 Lafarge	France	40.0	32.0	40.0	8.0	B
13	281 BHP Billiton	Australia	66.1	12.6	13.0	66.1	D
14	459 LVMH	France	36.0	26.0 ^z	36.0	32.0	G
15	4 BP	UK	36.3	48.1	36.3	na	B
16	15 TotalFinaElf	France	55.6	8.4	55.6	na	D
17	99 Suez	France	74.0	11.0	74.0	5.0	D
18	210 Ericsson	Sweden	46.0	13.2	46.0 ^m	25.9	B
19	394 Danone Groupe	France	60.3	na	60.3	na	D
20	123 Vodafone	UK	93.1	0.1 ^z	93.1	4.8	D

Sources: Individual Annual Reports.

Note: This table is constructed from the UNCTAD (2001) source which lists the world's largest 100 MNEs by foreign asset size. The foreign and total sales of these 100 MNEs are also reported so (F/T) sales ratios can be calculated. The Fortune Global 500, 2002, are included in this table. a. Japan; b. Europe; c. UK only; d. Europe, the Middle East and Africa; e. Japan; f. Europe; g. Global; h. Biregional; i. Home-region oriented; j. Host-region oriented.

Testing associations



Correlations

Bi-variate correlations

- Looking at the linear relationship between two variables (from -1 to 1).

Most common:

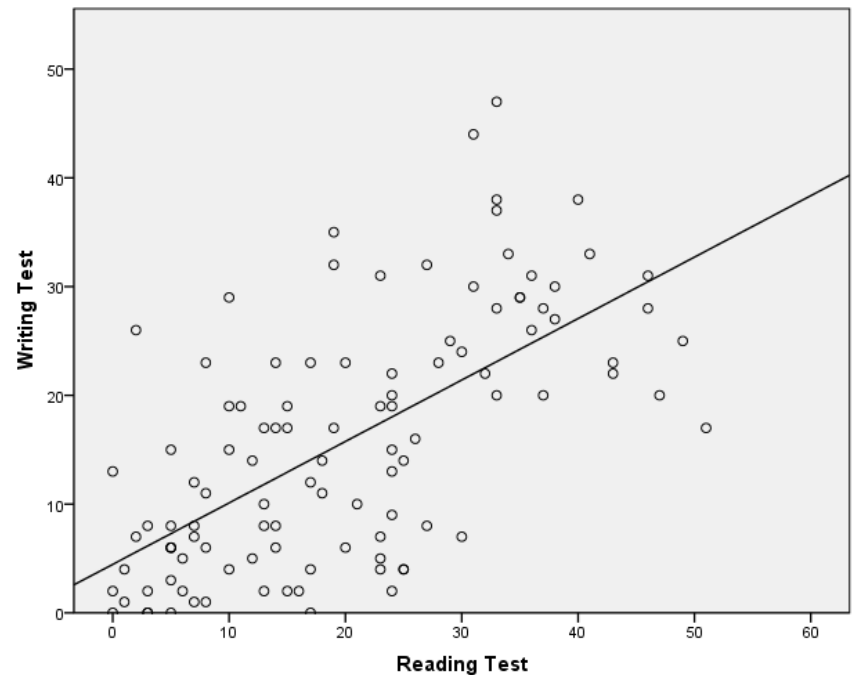
Pearson's correlation

- Random samples
- Both variables are interval/ratio

2nd most common:

Spearman's correlation

- Random samples
- The variables can at least be ordinal



Interpretation of coefficients



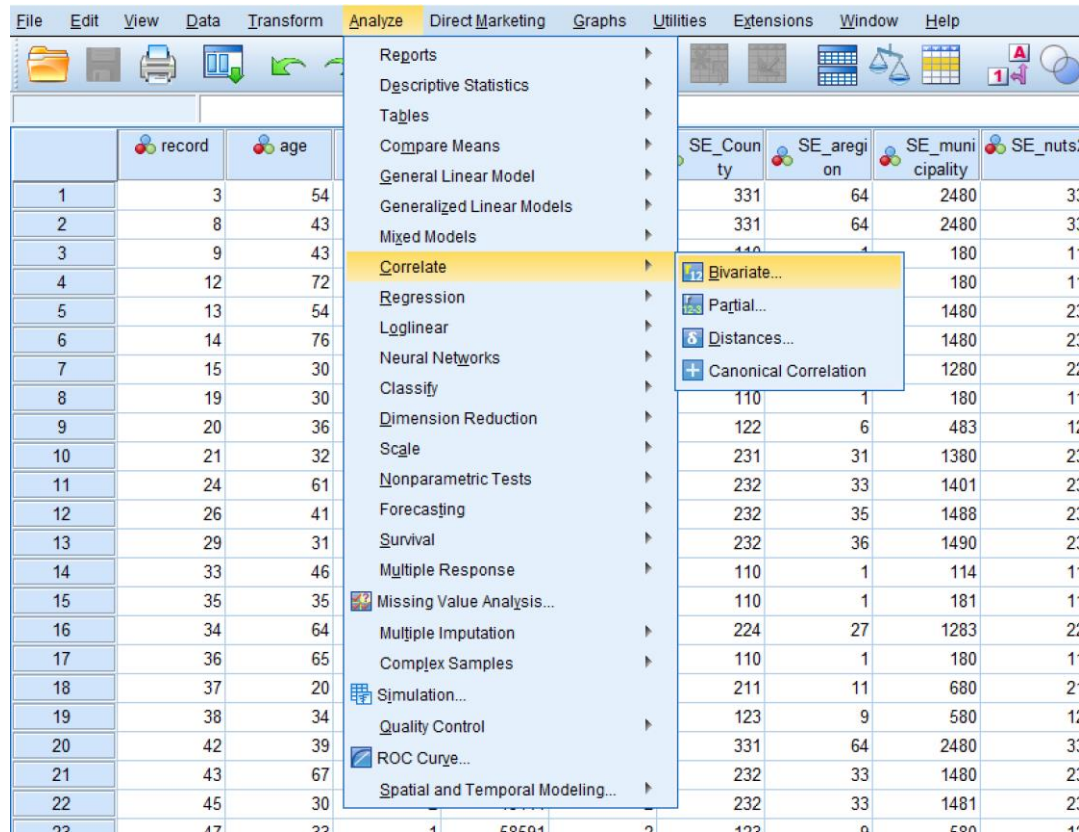
If Washington wins, the party of the incumbent president retains the presidency; if Washington loses, the opposition party will win.

A coefficient is not remarkable simply because it is statistically significant!

It must be practically meaningful.

Correlations

Analyze → Correlate →
Bivariate...



The screenshot shows the SPSS software interface. The 'Analyze' menu is open, and the 'Correlate' option is highlighted. A sub-menu is also open, showing 'Bivariate...' as the selected option. The background displays a data table with columns 'record' and 'age'.

	record	age
1	3	54
2	8	43
3	9	43
4	12	72
5	13	54
6	14	76
7	15	30
8	19	30
9	20	36
10	21	32
11	24	61
12	26	41
13	29	31
14	33	46
15	35	35
16	34	64
17	36	65
18	37	20
19	38	34
20	42	39
21	43	67
22	45	30
23	47	22

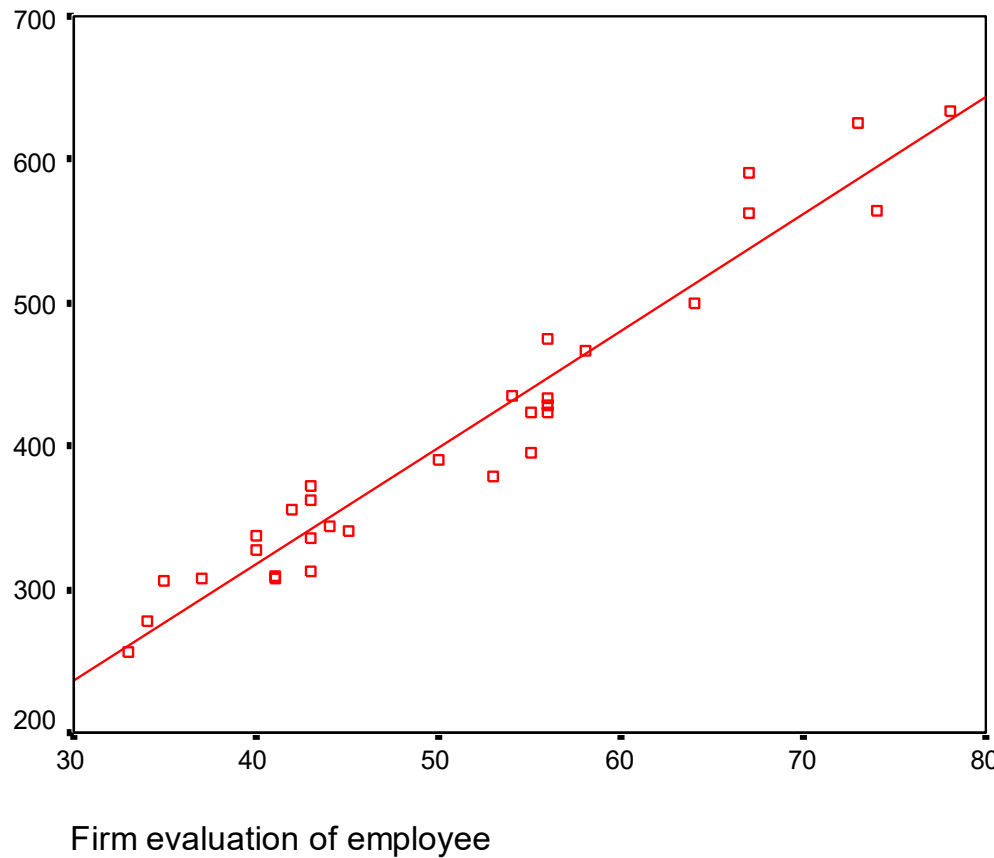
Correlations

Correlations

		Daily earnings	Seniority (in months)	Gender	Firm evaluation of employee
Daily earnings	Pearson Correlation	1	.900**	.169	.972**
	Sig. (2-tailed)	.	.000	.363	.000
	N	31	31	31	31
Seniority (in months)	Pearson Correlation	.900**	1	-.071	.849**
	Sig. (2-tailed)	.000	.	.703	.000
	N	31	31	31	31
Gender	Pearson Correlation	.169	-.071	1	.206
	Sig. (2-tailed)	.363	.703	.	.267
	N	31	31	31	31
Firm evaluation of employee	Pearson Correlation	.972**	.849**	.206	1
	Sig. (2-tailed)	.000	.000	.267	.
	N	31	31	31	31

** . Correlation is significant at the 0.01 level (2-tailed).

Correlations



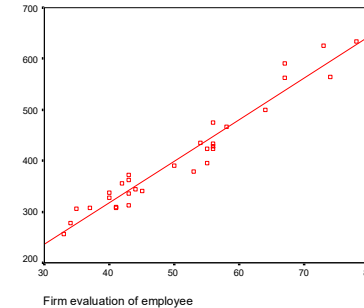
With a correlation coefficient of 0.972 one is not surprised that the data closely fit around a line.

To generate a graph like this: **GRAPHS** → **SCATTER** → **SIMPLE** → **DEFINE**

Regression analysis

If a **scatterplot** shows a linear relationship, we can summarize this pattern by drawing a line

A **regression** line summarizes the relationship between two variables in a specific setting: when one of the variables helps to explain (or predict the other)



Causality is the big difference between correlation and regression.

It is often a “theoretical causality”

The independent variable is assumed to influence the dependent variable.

Regression analysis

Consider the following example:

RQ: What drives earnings of employees in private companies?

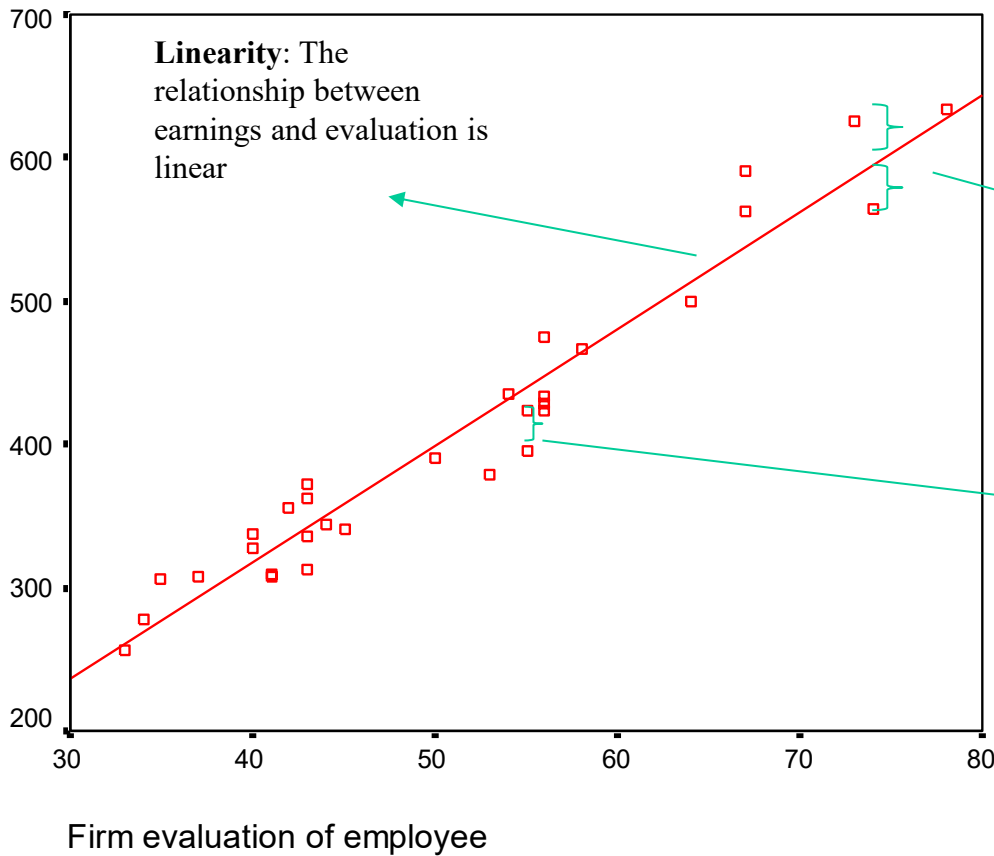
H: Seniority is positively related to earnings

A hypothesis is phrased on the population, not the sample

Theory provides us with other independent variables as well:

- Seniority
 - Gender
 - Evaluation
- EARNINGS

Assumptions

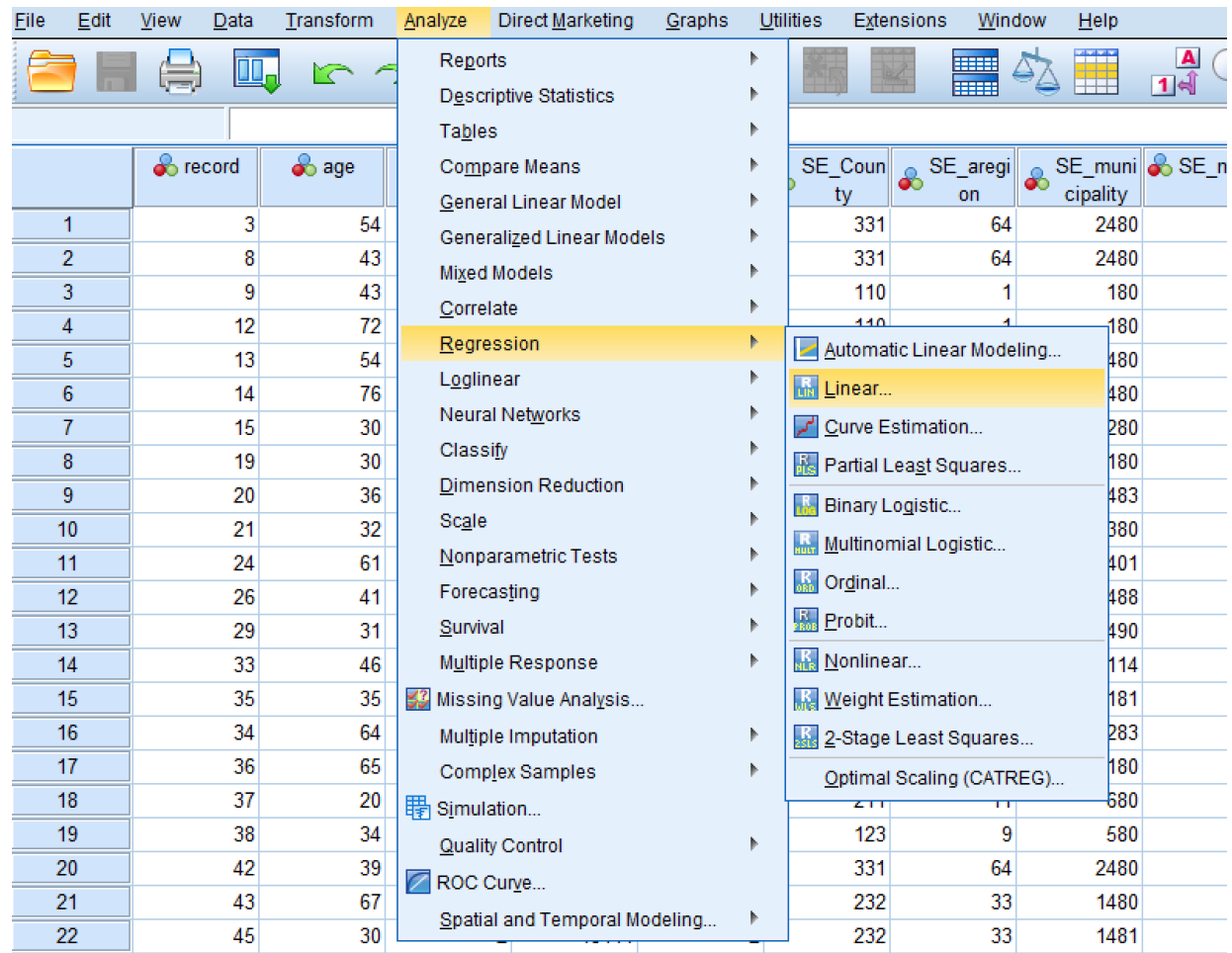


Independence: The independent variables are not correlated. They are in the prior example (evaluation and seniority, $\rho = 0.849$)

Regression analysis

ANALYZE → REGRESSION
→ LINEAR.

Specify your dependent and independent variables.



The screenshot shows the SPSS software interface. The 'Analyze' menu is open, and the 'Regression' option is highlighted. The 'Linear...' option is also highlighted in the submenu. The data view shows a table with columns 'record' and 'age'.

record	age
1	3
2	8
3	9
4	12
5	13
6	14
7	15
8	19
9	20
10	21
11	24
12	26
13	29
14	33
15	35
16	34
17	36
18	37
19	38
20	42
21	43
22	45

Regression analysis

R^2 – The coefficient of determination (explained variance)

Model Summary

Model	R	R Square	Adjusted R Square	Std. Error of the Estimate
1	.983 ^a	.966	.962	20.011

a. Predictors: (Constant), Firm evaluation of employee, Gender, Seniority (in months)

Refer to this when discussing the model.

Regression analysis

Evaluating the entire equation

ANOVA

Model		Sum of Squares	df	F	Sig.
1	Regression	309532.5	3	257.660	.000 ^a
	Residual	10811.886	27		
	Total	320344.4	30		

Small value (<0.05) indicates significance.

- a. Predictors: (Constant), Firm evaluation of employee, Gender, Seniority (in months)
b. Dependent Variable: Daily earnings

The F-statistics indicates the significance of the entire equation (and not the individual parts).

Regression analysis

Look for “big” error terms

Compare variables - use standardized effect sizes, otherwise unstandardized

Coefficients^a

Model		Unstandardized Coefficients		Standardized Coefficients	t	Sig.
		B	Std. Error	Beta		
1	(Constant)	70.344	24.746		2.843	.008
	Seniority (in months)	.288	.070	.311	4.089	.000
	Gender	9.769	8.355	.048	1.169	.253
	Firm evaluation of employee	5.858	.651	.698	9.000	.000

a. Dependent Variable: Daily earnings

Evaluating the parameters
(independent variables)

Small ($p < 0.05$) indicates significance

How it looks in research...

TABLE 1 Means, Standard Deviations, and Pearson Correlations

Variable	Mean	S.D.	1	2	3	4	5	6	7	8	9	10
1 Subsidiary location (Finland = 0, China = 1)	0.45	0.50										
2 Entry mode (Greenfield = 0, Acquisition = 1)	0.44	0.50	-0.46**									
3 Profitability	5.00	1.65	-0.06	-0.06								
4 Subsidiary size (log-transformed)	2.16	5.87	-0.28**	0.22**	0.03							
5 Importance of financial performance in performance evaluation	6.30	1.00	-0.24**	0.08	0.22**	0.07						
6 Inward transfer of business knowledge	11.28	3.95	0.03	-0.06	0.13+	-0.07	0.01					
7 Inward transfer of technical knowledge	6.75	3.74	0.37***	-0.12+	-0.04	0.13+	-0.07	0.3***				
8 Subsidiary external embeddedness	16.69	5.39	-0.09	0.00	0.15+	0.03	0.11+	-0.06	0.04			
9 Autonomy of strategic decision-making	35.19	8.45	0.05	0.21**	-0.05	0.16*	0.08	0.02	0.20**	0.12+		
10 Importance of knowledge development in performance evaluation	4.73	1.51	0.13+	0.00	0.11+	-0.05	-0.024	0.11+	-0.03	0.20**	0.24**	
11 Knowledge development	19.28	5.47	0.06	0.03	0.13+	0.09	0.05	0.10	0.41***	0.33***	0.31***	0.24**

+ p < .10, * p < .05, ** p < 0.01, *** p < .001
All one-tailed tests.

TABLE 2 Regression Models

	Model 1: Control only				Model 2: Full model			
	Beta	t	Sig.	VIF	Beta	t	Sig.	VIF
Control Variables								
Subsidiary location (Finland = 0, China = 1)	0.144	1.543	0.125	1.397	-0.09	-0.992	0.323	1.812
Entry mode (Greenfield = 0, Acquisition = 1)	0.075	0.831	0.407	1.306	0.07	0.085	0.932	1.401
Profitability	0.130	1.600	0.112	1.06	0.112	1.572	0.118	1.122
Subsidiary size (log)	0.108	1.308	0.193	1.096	-0.027	-0.363	0.717	1.221
Importance of financial performance in performance evaluation	0.040	0.487	0.627	1.108	-0.006	-0.080	0.936	1.127
Independent variables								
Inward transfer of business knowledge					-0.041	-0.553	0.581	1.205
Inward transfer of technical knowledge					0.422***	5.144	0.000	1.500
Subsidiary external embeddedness					0.246**	3.453	0.001	1.126
Autonomy of decision-making					0.189*	2.541	0.012	1.233
Importance of knowledge development in performance evaluation					0.086	1.17	0.244	1.193
Model statistics								
R	0.199				0.575			
R square	0.040				0.33			
Adjusted R square	0.009				0.285			
R square change	0.040				0.291***			
F	1.275				7.352			
F change	1.275				12.935***			
N	161				161			

+ p < .10, * p < .05, ** p < 0.01, *** p < .001
All one-tailed tests.

Data in the table present standardized beta coefficients.

Dependent Variable

Subsidiary Knowledge Development

To measure the development of subsidiary knowledge, we asked respondents to “rate the extent to which the subsidiary has developed knowledge that is superior compared to that of other units in the business area during the last three years.” Respondents answered this question separately for each of the following functions: general management, manufacturing, marketing/sales, service, and research and development (R&D).² We used a seven-point Likert scale where 1 = “very much lower” and 7 = “very much higher” compared with other subsidiaries. The sum of these questions represents the overall level of subsidiary knowledge development compared to other MNC subsidiaries.

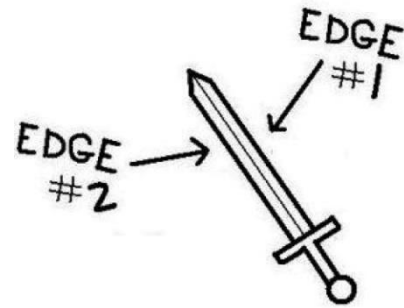
Results

We used multiple regression analysis as the analysis method in our study because it provides a means for objectively assessing the degree and character of the relationship between dependent and independent variables by forming the variate of independent variables (Hair, Anderson, Tatham, & Black, 1998). Regression analysis, thus, enabled us to assess the magnitude and direction (positive or negative) of our hypothesized relationships. The statistical program used was SPSS 12.0.

Regression analysis

A model with a lot of variables

- Overly complex
- Loss of precision



A model with few variables

- The estimates become biased (omitted variable bias)
 - Hypothesis tests and significance levels can be misleading



Next session: Tomorrow, 10.15-12. Quantitative analysis using case studies.